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DIGITAL DESIGN FOR A SWEEPING LANGMUIR PROBE BOARD

by

Omar Morales

A report submitted in partial fulfillment
of the requirements for the degree

of

MASTER OF SCIENCE

in

Electrical Engineering

Approved:

Dr. Charles Swenson
Major Professor

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Committee Member

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UTAH STATE UNIVERSITY
Logan, Utah

2004

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Abstract

Digital Design For A Sweeping Langmuir Probe Board

by

Omar Morales, Master of Science

Utah State University, 2004

Major Professor: Dr. Charles Swenson
Department: Electrical and Computer Engineering

In-situ probing of the Earth's upper atmosphere has greatly contributed to the understanding of the ionosphere's variations and structures. The purpose of this research project has been to develop the digital component of a Swept Langmuir Probe that will be used on a sounding rocket to be launched through the equatorial ionosphere. The probe will make measurements of the plasma density and temperature in the ionosphere.

The digital component uses an FPGA to perform three main functions in parallel which are generating a sweeping voltage on the probe, sampling data channels, and communicating measured data to external devices. The FPGA controls a Digital to Analog converter to create a sweeping voltage. The current measured from the probe is split into two different channels with different gains to allow for a more dynamic range of current measurements. Sampling accomplished by controlling Analog to Digital converters and co-adding the results eight times to reduce noise. The FPGA provides two different interfaces for communication to the outside world. The interfaces are a serial communication interface, via RS422 transceivers with a NASA telemetry encoder, and an interface with a PC for calibration purposes. The purpose of calibration is to measure how the instrument works under varying conditions to help understand the effect the instrument has on the readings. To accomplish this task a special diagnostic mode was programmed into the

FPGA that allows a PC to control the probe. The PC is able to control the voltage sweep and decide when to sample a desired channel.

A top down approach is taken to simplify the design by partitioning the design into smaller and easier pieces to work with. This report will give an overview of how the instrument functions, how it was designed, how it was programmed using VHDL, and then show results of how it worked.

(70 pages)

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I would also to thank the other students that worked on this project, for their team effort in completing it. Namely: Tim Campbell, for making the Calibration interface GUI; Albert Hummel, for designing the analog components and for writing Matlab scripts to interpret the measured data; Hazen Sedgwick, for taking care of all of the mechanical needs for the probe; and Ken VanHille, for getting the project started and laid out.

Omar Morales

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Chapter 1
Introduction

The ionosphere near the equator presents an interesting problem for satellite communications. Typically high frequency radio waves pass through the equatorial ionosphere with out distortion. At other times these radio waves are severely scattered by density variations within the ionosphere. The resulting scintillations on the received radio signals can at times block normal operations. The location of these scattering sources in the ionosphere has been studied by radar. One source of disturbances is a layer at about 180 Km. This has been observed for years and often occurs as a precursor to even larger scattering from all altitudes in the range of 200 to 1000 Km. This phenomenon is known as equatorial spread F (ESF). ESF is when the ionosphere becomes unstable and produces large turbulent variations of electron density at F region heights producing large index of refraction variations for radio waves traversing the region. The scattering leads to RF scintillation effects which interfere with communication links such as HF communications, point to point communications, satellite to ground communications, GPS, and over-the-horizon radars [1].

A NASA research program titled "Rocket Investigation of Scattering Layers in The Bottom Side Equatorial F-Region Ionosphere," also known as "The Spread-F Trigger Program," is aimed at investigating the electrodynamics of these lower altitude scattering layers that are precursors to the more important Spread-F. Sounding rockets will make measurements at altitudes from about 90 to 200 km of ionospheric densities, temperatures, collision frequencies, electric and magnetic fields, and winds. The goal is to understand the radio wave scattering phenomena at 180 Km which seems to be a trigger for larger ionospheric instability. The ultimate goal of the investigation is to have improved ability to predict satellite communication outages caused by Spread-F disruptions.

1.1 Program Overview

1.1.1 Instruments

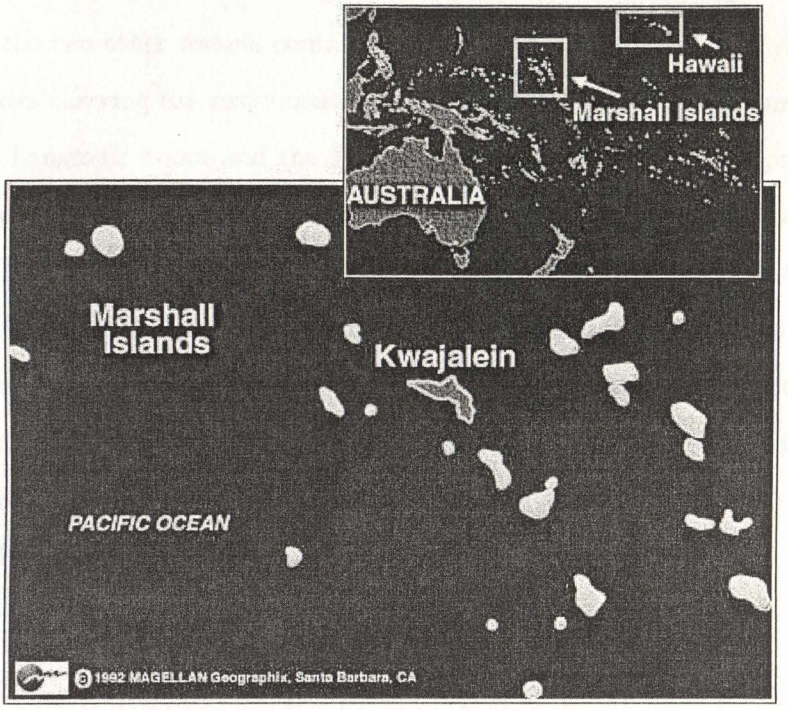

Fig. 1.1: Map of location of Kwajalein.

The sounding rockets of the Spread-F Trigger program will be launched from the Roi Namur range in the Kwajalein Atoll as shown in fig. 1.1. Four universities are contributing to the program. Dr. David Hysell from Cornell is the principle investigator. Dr. Miguel Larsen from Clemson University is responsible for chemical release Trimethyl-Aluminum (TMA) payloads. The TMA payloads will be used to measure the vector wind profiles through photographic triangulation of the TMA puff along the rocket trajectory. For the instrumented rockets, Penn State is responsible for an electric field detector, and Utah State University is responsible for a Plasma Impedance Probe (PIP) and a Sweeping Langmuir Probe (SLP). The PIP includes a plasma frequency probe, a plasma-sweeping probe, and a DC probe. The Utah State probes provide plasma density, temperature, and collision frequency measurements necessary for estimating the ionospheric conductivity.

1.1.2 Rockets

The program consists of a total of six rockets being launched on two different nights. Each set of launches will consist of three rockets, one rocket containing an instrumented payload and the two other rockets containing TMA payloads.

The rockets carrying the instrumented payloads will eject their noise cones and skirts, exposing the Langmuir boom and the PFP monopole boom. Figure 1.2 presents a mechanical drawings of the instrumented payloads before and after the noise cone and skirt have been ejected. The Utah State University Langmuir sensor consists of a short 2.375 in. section at the tip of a 20.25 in. boom that has a .875 in. diameter. This boom is mounted on the forward deck under the noise cone. The plasma impedance probe consists of a 32 in. monopole antenna with a 1.0 in. diameter mounted on the aft end of the payload. The payload is ejected away from the motors at an altitude of 98.2 Km and the nose cone at an altitude of 90.7 Km.

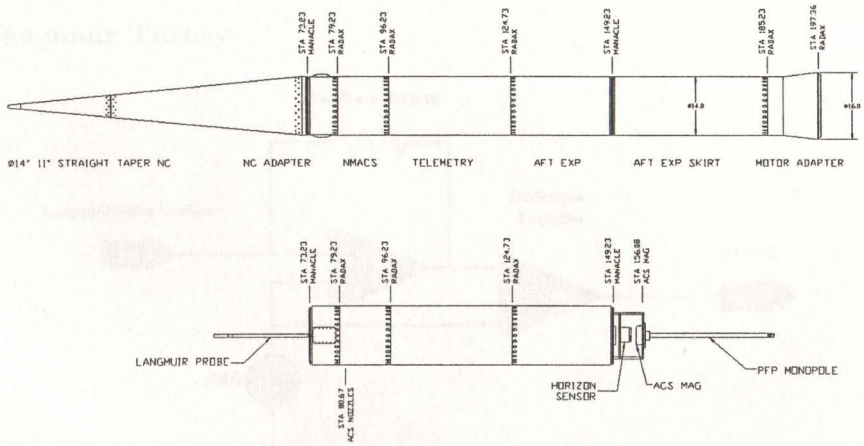

Fig. 1.2: Two views at the forward deck showing the sensor boom and the electronics box mounted on the deck just underneath.

Figure 1.3 is a drawing of the Sweeping Langmuir Probe mounted on the rockets deck. It shows how the electronics enclosure and the boom are mounted together on the deck of the rocket. The PFP is connected in a similar manner. This thesis will report on the

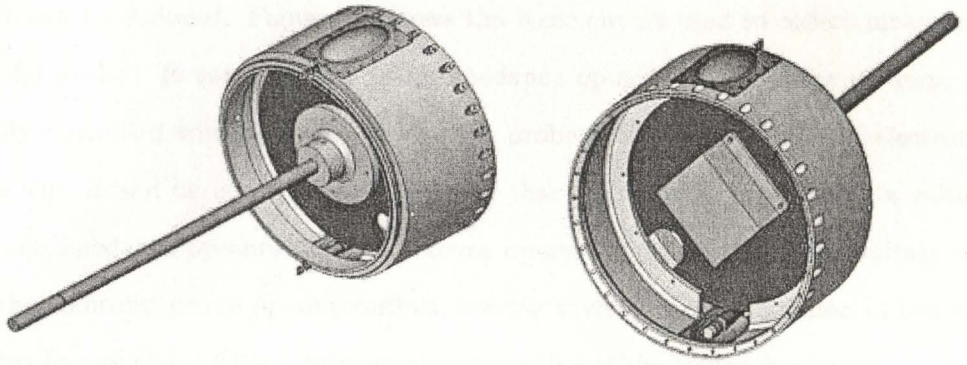

Fig. 1.3: The Langmuir boom and electronics are mounted on the deck of the rocket.

interface and control electronics of the Utah State University Sweeping Langmuir Probe (SLP) [2], [3].

1.2 Sweeping Langmuir Probe Board Overview

1.2.1 Langmuir Theory

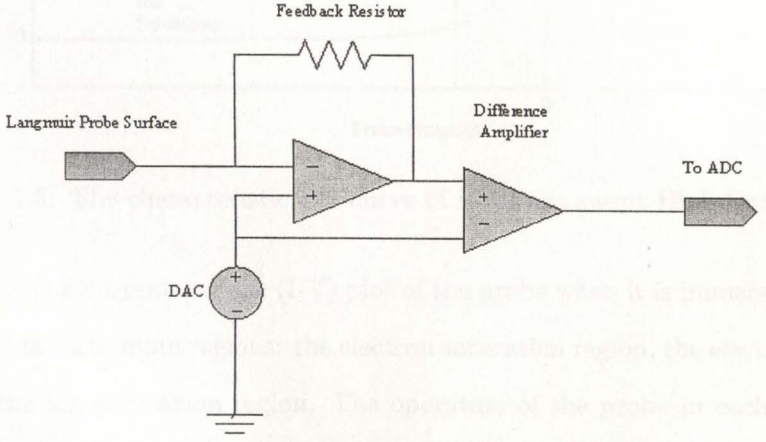

Fig. 1.4: A basic circuit used to measure current on a Langmuir Probe.

Swept Langmuir Probes have been flown before on sounding rockets, and are the most widely used probes to measure plasma density and temperature. The probe measures the current that can be drawn from the sensor for a given applied voltage. From the current

voltage relationship of the sensor immersed in a plasma, the density and temperature of the plasma can be deduced. Figure 1.4 shows the basic circuit used to collect measurements from the probe. It consists of a transimpedance op-amp, a difference op-amp, and a digitally controlled voltage source. When the probe is immersed in plasma electrons will collide with it and be absorbed. The current that is drawn is converted to a voltage by the transimpedance op-amp. The difference op-amp then subtracts the voltage supply from the transimpedance op-amp output, leaving a voltage representation of the current collected by the probe with a gain equal to the value of the feedback resistor.

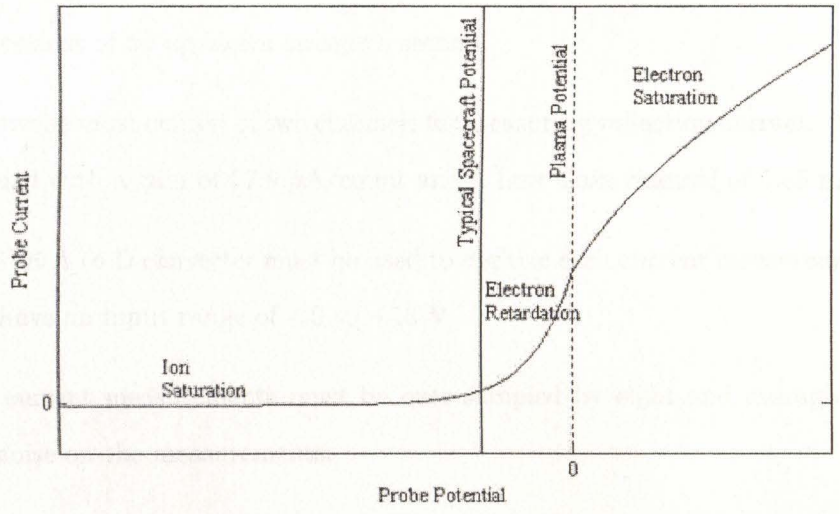

Fig. 1.5: The characteristic I-V curve of a voltage swept High Gainrobe.

Figure 1.5 is a current-voltage (I-V) plot of the probe when it is immersed in a plasma. The plot depicts three main regions: the electron saturation region, the electron retardation region, and the ion saturation region. The operation of the probe in each of the regions allow for the measurements of the plasma temperature and density. A sweeping voltage ranging from -1V to 5V will be applied to the USU Langmuir Probe in the form of a triangle wave. While in the ion saturation region information about the ion density can be obtained. While in the electron retardation region the temperature of the plasma can be obtained from the slope of ln(I). In other words, the electron temperatures determine

the length of the electron retardation region. For positive voltages the probe operates in the electron saturation region and the electron density can be determined [4], [5].

1.2.2 Sweeping Langmuir Probe Board Requirements

The design of the USU Sweeping Langmuir Probe began in March 2003 and started by determining the probe requirements. These requirements are:

1. To measure the current collected due to an applied voltage which would sweep over the range of -1 to 5 V.

2. The voltage sweep must be linear up from -1 to 5 V and linear down from 5 to -1 V, and consist of 50 up/down sweeps a second.

3. The probe must consist of two channels for measuring collection current. A High Gain channel with a gain of 57.9 pA/count and a Low Gain channel of 5.85 nA/count.

4. A 16-bit A to D converter must be used to digitize each current measurement channel and have an input range of -10 to +10 V.

5. The current measurements must be over sampled by eight and averaged to reduce the noise on the measurements.

6. Support a diagnostic calibration mode were the voltage step on the probe can be manually controlled.

7. Key housekeeping voltages and signals must be monitored to ensure instrument health.

8. Output data must be output in an RS422 serial link to NASA telemetry equipment for down-link.

The Sweeping Langmuir Probe was also required to interface to an existing power board and a Low Cost Telemetry Encoder (LCTE) which SDL had previously designed and built [6]. The SLP board will use the already designed power board for this mission

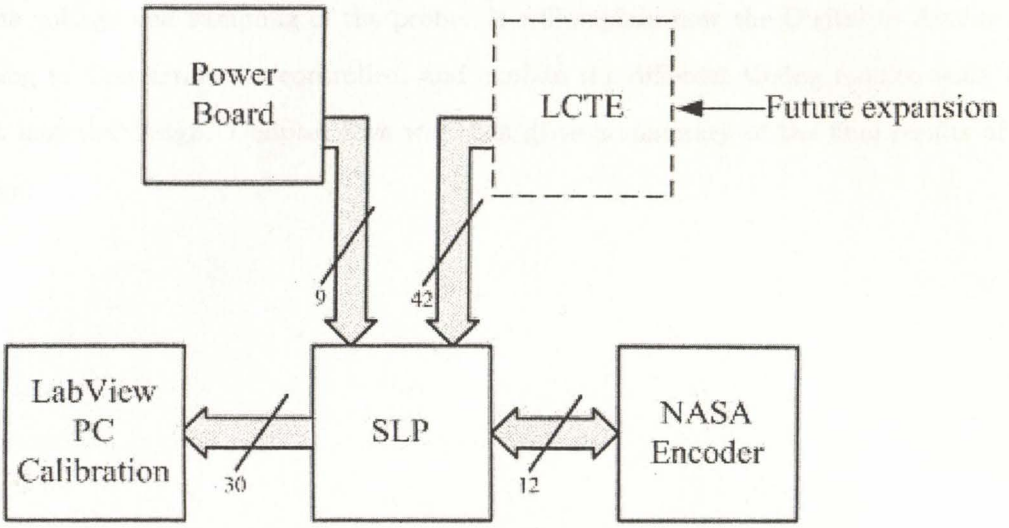

Fig. 1.6: SPL interface diagram.

however, it will not be connected with the LCTE for this program. It was desired that the SLP board have the capability to interface with the already functioning and adaptable LCTE board to reduce future design costs. Figure 1.6 is a block diagram showing how the SLP board is to interface with other systems. A backplane connector will interface it with the power board and the LCTE. Two other connectors will provide the interfaces for calibration with a PC for serial communication with NASA's PCM encoder.

1.3 Description of Thesis

This thesis presents the digital design of the Utah State University Sweeping Langmuir Probe (SLP). The digital component of the the probe was required to control the voltage sweep of the probe and control the sampling of the current measurements. It also provided a data handling function by sampling various housekeeping data for the instrument and providing this data in a serial format to NASA's telemetry system. The digital component provides an interface with a PC for calibration and manual operation of the instrument. Chapter Two presents a top-level design overview of the SLP. Chapter Three presents the Calibration and NASA interfaces. It will give a detailed description of how the interfaces work, and then it will show results from testing each of the interfaces. Chapter Four will describe how the digital component of the probe was designed to control the sweeping

of the voltage and sampling of the probe. It will explain how the Digital to Analog and Analog to Converters are controlled, and explain the different timing requirements that went into the design. Chapter Five will then gives a summary of the final results of the design.

Chapter 2
Top-Level Design Overview

This chapter will give a broad overview of how the instrument functions and the characteristics of the channels that are measured by the electronics. It will describe the main electronics that are used by the instrument and how the FPGA controls them. The chapter will also describe the different communication interfaces supported by the probe.

2.1 Instrument Overview

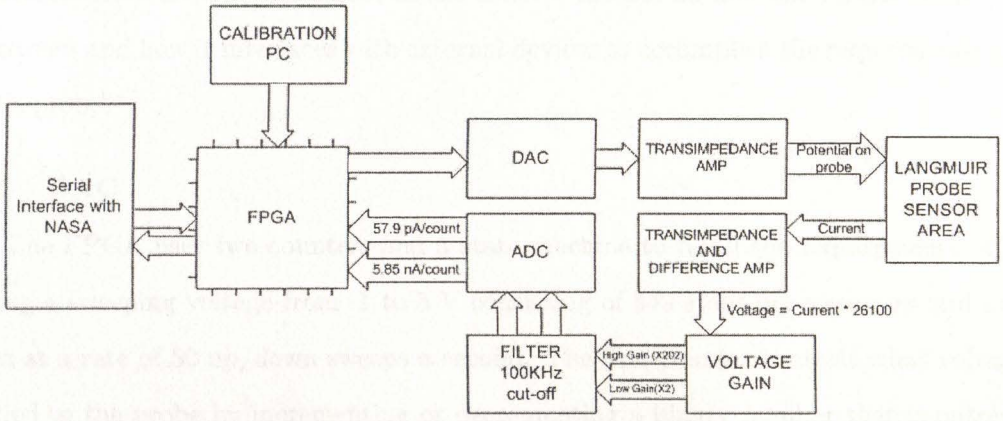

Fig. 2.1: High Gainrobe data flow.

Figure 2.1 shows a basic diagram of the electronics that go into the Langmuir Probe [7]. The instrument uses an FPGA to provide two interfaces for communication to the outside world. The two interfaces are a serial communication interface with a NASA telemetry encoder, and an interface with a PC for calibration purposes. Each interface requires its own connector on the board. In addition to communication to external sources the FPGA controls a Digital to Analog converter (DAC) to create a sweeping voltage. This voltage is applied to the probe via a transimpedance amplifier. The current measured from the probe is split into two different channels called a Low Gain and a High Gain channel. The Low Gain channel contains a voltage representation of the current with a gain of 52,200

V/A, and the High Gain channel is similar but with a gain of 5,272,200 V/A. To diminish the noise arriving at the Analog to Digital converters (ADC) these two channels are low pass filtered with a cut-off frequency of 100 KHz. Two ±10 V input 16 bit ADCs are used to sample both of the channels. The output of the ADC for the Low Gain channel has a resolution of 5.85 nA/count with a range ±191.6 μA. The High Gain channel has a resolution of 57.9 pA/count with a range of ±1.897 μA.

2.2 FPGA Overview

The FPGA has three main functions that it needs to perform in parallel which are: generating a sweeping voltage, sampling data channels, and communicating measured data to external devices. The remainder of the chapter focuses on how the FPGA controls the electronics and how it interfaces with external devices to accomplish the requirements given to the probe.

2.2.1 DAC

The FPGA uses two counters and a state machine to fulfill the requirements of generating a sweeping voltage from -1 to 5 V consisting of 549 steps per sweep up and sweep down at a rate of 50 up/down sweeps a second. The first counter controls what voltage is applied to the probe by incrementing or decrementing a binary number that is output to the DAC. The second counter keeps track of how many steps are taken in a sweep. When the maximum number of steps are taken it toggles the first counter to change direction of the count and thus the sweep. The state machine controls when the counters should increment and when data is written to the DAC. This allows for the sweep rate to be maintained at 50 Hz.

2.2.2 ADC

Three 16-bit ADCs are used to sample the High Gain, Low Gain, and Housekeeping channels. Figure 2.2 shows the housekeeping signals that are monitored and how the ADCs interface with the FPGA [7]. The electronics on the probe require seven different voltage source levels to operate. All these voltages are monitored by the housekeeping channel.

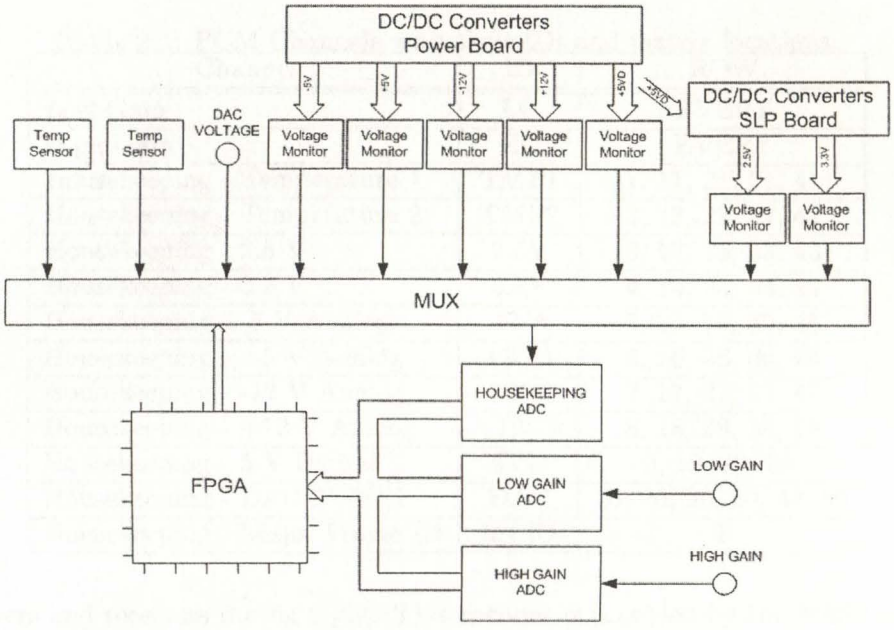

Fig. 2.2: Housekeeping electronics diagram.

Since the ADC can only support voltage ranges from ± 10 V, some of the voltages are scaled to be within the operating range of the ADC. Two temperature sensors, which are located on opposite corners of the electronics board, are monitored to help keep track of the temperature drift that the probe may encounter when in flight. The voltage that the DAC outputs is also monitored to ensure that a voltage sweep is occurring. With the help of a multiplexer the FPGA can select which housekeeping signal it wants to sample.

The FPGA over samples by eight and averages the data coming from the Low and High Gain channels. This is to fulfill one of the requirements to reduce noise from the system. A state machine controls when each ADC is sampled and a counter lets the state machine know when the over sampling is completed. Figure 2.2 shows that the ADCs share a data bus to the FPGA; the output bus from each ADC is held in high impedance unless the chip is selected.

2.2.3 NASA Serial Interface

The FPGA is required to support two communication interfaces with outside sources. The main interface transmits the measured samples to a WFF93 PCM encoder via RS422

Table 2.1: PCM Channels with their IDs and matrix locations.

Channel	ID	ROW
Low Gain	LG	EVERY
High Gain	HG	EVERY
Housekeeping - Temperature 1	TMP1	1, 11, 21, 31, 41
Housekeeping - Temperature 2	TMP2	2, 12, 22, 32, 42
Housekeeping - 2.5 V	2.5V	3, 13, 23, 33, 43
Housekeeping - 3.3 V	3.3V	4, 14, 24, 34, 44
Housekeeping - -5 V Analog	-5VA	5, 15, 25, 35, 45
Housekeeping - +5 V Analog	+5VA	6, 16, 26, 36, 46
Housekeeping - -12 V Analog	-12VA	7, 17, 27, 37, 47
Housekeeping - +12 V Analog	+12VA	8, 18, 28, 38, 48
Housekeeping - 5 V Digital	5VD	9, 19, 29, 39,
Housekeeping - DAC SWEEP	DAC	10, 20, 30, 40, 49, 50
Housekeeping - Major Frame ID	MFID	1

transceivers and receivers during flight. This encoder is provided by the NASA Sounding Rocket Operations Contract (NSROC). The WFF93 PCM system is encoder hardware used to support the NASA Sounding Rocket Program. It serves as a general-purpose reconfigurable telemetry encoder for these types of programs. The WFF93 PCM system has different modules that can support analog, serial, and parallel inputs [8]. The Langmuir Probe will interface with the serial deck of the encoder. This interface is also referred to as the RS422 interface, since RS422 differential signals are used to communicate with the serial deck.

Pulse Code Modulation (PCM) is used to combine the multiple channels of data into a single stream of data. Figure 2.3 illustrates the PCM matrix used for the program. Each row in the matrix is called a minor frame and each matrix is called a major frame. Frame synchronization words are used to the decode the data contained in the matrix. In this matrix the Frame synch words are located in the last two columns of the matrix. Table 2.1 lists the channels that are contained in the PCM matrix, and defines their reference IDs, and their locations in the matrix.

The PCM encoder provides five differential handshaking signals that control communication with the probe. Two of the signals are for synchronization with the PCM matrix. The encoder will assert a strobe signal called Major Frame at the end of each major frame

in the matrix. In this case the signal Major Frame will be asserted when the matrix is at row 50 column 50. It will also assert a strobe signal called Minor Frame when the matrix reaches column 50, which is the end of a minor frame. These two strobes help keep the High Gain in synch with the PCM encoder. The synchronization with the encoder is necessary because data measured from an entire voltage sweep of the probe must fit inside a single major frame. The other handshaking signals are called Load, Gated Clock, and Enable. The Load signal indicates when a new word should be loaded into a shift register to be output serially. The Gated Clock controls when each bit is output from the shift register. In this program the Gated Clock will run at 2 Mbits/sec. The signal Enable enables the shift register. If the signal Enable is not asserted then no communication should take place regardless of what the other signals do.

2.2.4 Calibration

The second required interface is one that will allow for calibration of the instrument. The purpose of calibration is to measure how the instrument works under varying conditions to help understand the effect the instrument has on the readings. To accomplish this task, a special diagnostic mode was programmed into the FPGA that allows a PC to control the probe. The PC is able to specify when a voltage step occurs, and when to sample a desired channel. The only housekeeping channels available during calibration are the the DAC voltage output and the two temperature sensors. A GUI made from the program LabView was developed to communicate with the probe. The GUI allows the user to specify what voltage level to sweep to and read the outputs from the Low and High Gain channels at the specified voltage. It also allows for an entire sweep to be run while simultaneously recording data received from the probe and data measured from an external multimeter. The two sets of measured data aid in calculating the noise in the system.

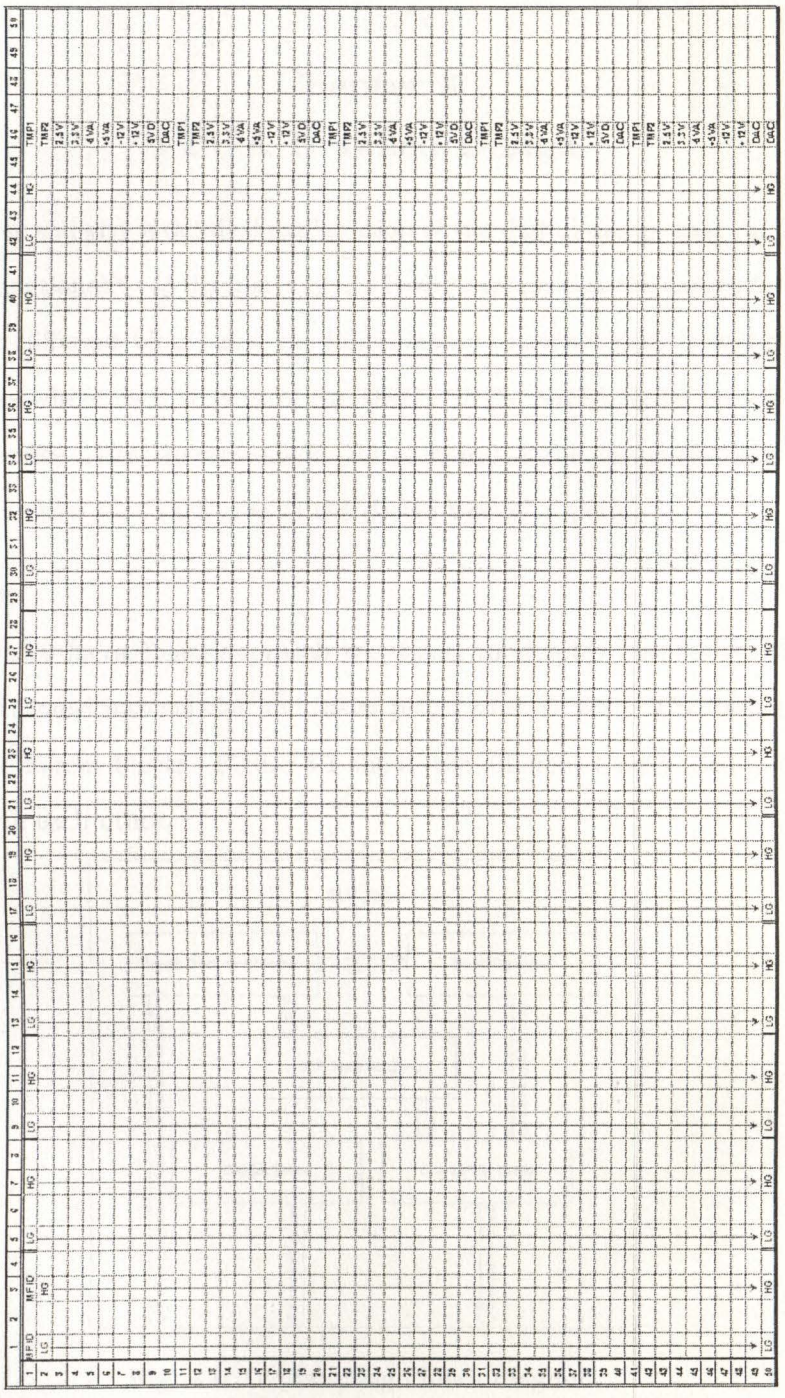

Fig. 2.3: PCM matrix.

Chapter 3
FPGA Detailed Design

This chapter will give a detailed explanation of the digital design that the FPGA uses to control the functionality of the instrument. It will explain how the design for the FPGA was broken into different blocks and the functionality of each block. It will then explain how these blocks where used to control the DAC, ADCs, NASA Serial interface, and the Calibration interface. After an explanation for each area of the design the results obtained from testing will show that each area of the design works properly.

3.1 Overview

A top down or hierarchical approach was taken to simplify the design. That is the system was partitioned into smaller and easier pieces to work with. The design was partitioned into six different blocks. Figure 3.1 shows the different blocks used and how they interconnect with each other.

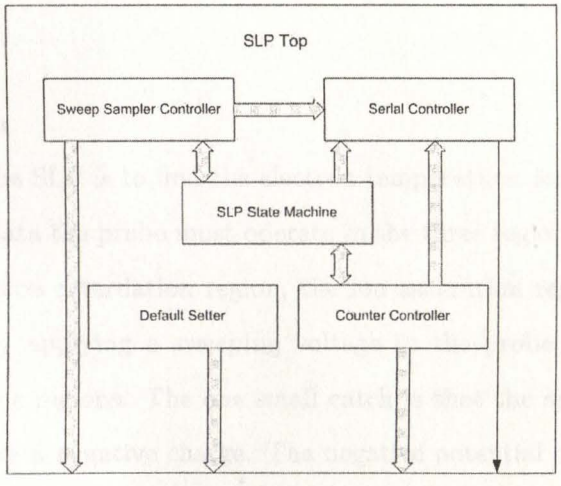

Fig. 3.1: Digital design block diagram.

The top-level block is called SLP Top. It simply defines the I/O on the FPGA and interconnects the five other blocks together. The block SLP State Machine can be considered

the backbone of the digital design. It loops through a series of states that send out control signals to each of the different blocks letting them know that they need to increment a counter, get a sample, write a sample to the FIFO, and so on. SLP State Machine also controls the different functionality for the board when it is in either Calibration or RS422 mode. The Counters Controller block contains several counters that act as timers and are used to regulate all of the timing requirements on the board, such as the 50 Hz sampling rate and the 10 μs settling time for the DAC. The Sweep Sampler Controller could actually be split up into two different blocks, but a design decision was made to maintain it as it is. Its primary functions are to generate binary values that are output to the DAC that in turn output the swept voltage to the boom; it also samples the High and Low gain channels eight times and averages the samples before they get sent out to the telemetry encoder. The Serial Controller stores all of the sampled data into a FIFO and then outputs the data serially to a telemetry encoder. The Default Setter is a very simple block. It outputs reset signals to different chips on the board and it also selects the different modes that those chips are to operate in. Since we don't change the modes of any chips during operation of the board, it just sends out constant high or low signals to the different peripheral chips.

3.2 DAC

3.2.1 DAC Design

The purpose of the SLP is to find the electron temperature, ion density, and electron density. To find this data the probe must operate in the three regions described in chapter 1, which are the electron retardation region, the ion saturation region, and the electron saturation region. By applying a sweeping voltage to the probe it could be forced to sweep through all three regions. The one small catch is that the spacecraft, while in the ionosphere, will acquire a negative charge. The negative potential of the spacecraft could offset the sweeping voltage enough to cause the probe not to operate within all three regions. For this reason a voltage sweep from -1 to 5 V is used. This will help ensure that, even with a negative offset of a few volts, the probe will be able to operate in all three regions.

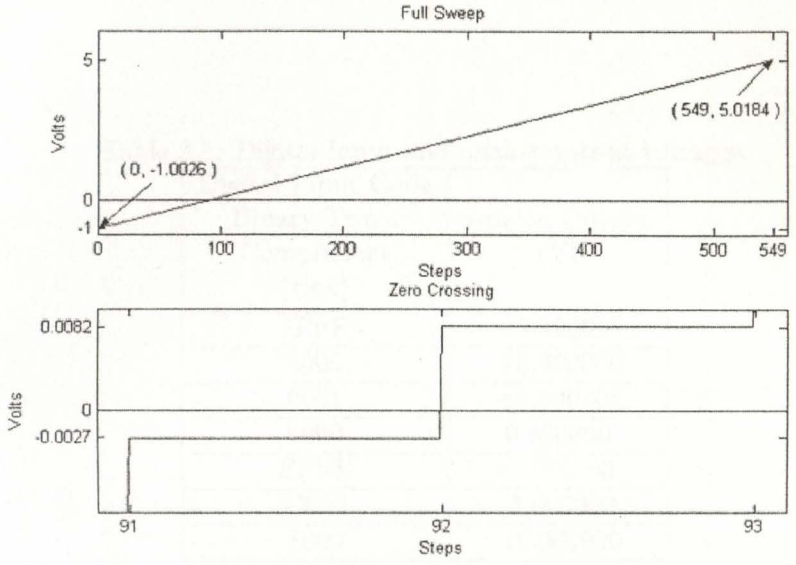

Fig. 3.2: SLP voltage range.

Figure 3.2 is a plot of the desired sweep. The sweep is to run at 50 Hz, that is sweep up to 5 V and sweep down to -1 V 25 times each per second. This will be done with a timer that regulates how often a voltage step is taken, and with a counter that adjusts a binary number by a fixed value or step. The binary number is called DAC_SWEEP in the design because it is the value that gets output to the DAC to generate the sweeping voltage. The DAC has a range of -10 to 10 V and converts twos complement binary numbers into a voltage. Table 3.1 illustrates voltage conversion codes for the DAC. Since the DAC does not have infinite precision we will not be able to achieve the exact range desired, but that not pose a problem as long as we know what voltages are being applied to the probe. The starting hex value used in the design is F32A which converts into -1.026 V. The DAC will be incremented by a step size of 36 counts or 10.98 mV. After 549 steps the ending hex value will be 403A, which converts to 5.0018 V.

The blocks that are used to control the DAC are the SLP State Machine, the Counters Controller, and the Sweep Sampler Controller. A little background to the state machine needs to be given before it is shown how the blocks control the DAC.

Table 3.1: Digital input and analog output voltages.

Digital Input Code Binary Two's Compliment (Hex)	Analog Output (V)
7FFF	+9.999695
4000	+5.000000
0001	+0.000305
0000	0.000000
FFFF	-0.000305
C000	-5.000000
8000	-10.000000

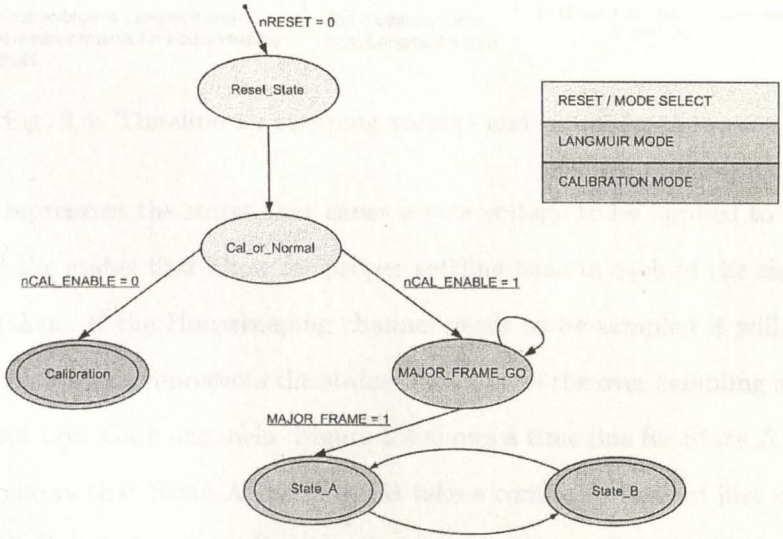

Fig. 3.3: Basic SLP State Machine diagram.

Figure 3.3 shows that the state machine has two main modes of operation. The Calibration Mode and the Langmuir Mode. Right now we will focus on how the DAC is controlled during Langmuir Mode. The Calibration Mode also controls the DAC but since it controls it in a similar manner as done in the Langmuir Mode we will only talk about the Calibration Mode as a whole later in the chapter. The Langmuir Mode contains the states that control what the instrument will do when operating in flight. There are three main states. The state MAJOR_FRAME_GO is used to get the FPGA in synch with the telemetry encoder that it communicates with. This way the FPGA starts sweeping the voltage and sampling channels at the beginning of a major frame. The states State_A and State_B aren't really states but representations of a group of states that perform a specific task.

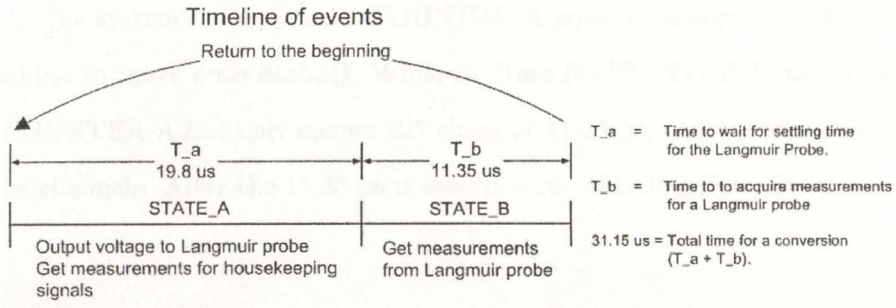

Fig. 3.4: Timeline for stepping voltage and sampling the probe.

State_A represents the states that cause a new voltage to be applied to the probe via the DAC and the states that allow for proper settling time in each of the channels before samples are taken. If the Housekeeping channel needs to be sampled it will also be done during State_A. State_B represents the states that control the over sampling and averaging of the High and Low Gain channels. Figure 3.4 shows a time line for State_A and State_B. The timeline shows that State_A and State_B take a combined time of just under 32 μs.

Figure 3.5 shows the state diagram that takes place in State_A. State_A starts off by checking for a stall which happens just before every Major and Minor Frame for synchronization purposes. If stalls aren't required in State_A then the state machine starts running through states that enable the DAC to latch and convert a new binary value that

was output by the Sweep Sampler Controller block. Once the value has been latched into the DAC an increment signal is sent to the Sweep Sampler Controller block to output the next binary voltage step value. The output of step voltages function as a post increment, because the current value is latched into the DAC; and then the counter is incremented for the next time. Sweep Sampler Controller also keeps track of the current position of the sweep. When the end of the sweep is reached it toggles the counter to start stepping the opposite direction. The sweep rate is controlled by two counters in Counters Controller called Counter A and Counter B. These counters are used as timers, and they combine to keep the overall sweep rate at 50 Hz. While the state machine is in State_A it continually enables COUNTER A, this allows the interpretation of each count to be 50 ns since the system clock runs at 20 MHz. After 396 counts or 19.8 μs, which is the settling time required by the system's components, COUNTER A asserts the signal MAX_A telling the state machine to move onto State_B. While in State_B COUNTER B works exactly the same as COUNTER A but only counts 227 times or 11.35 μs, which is the time needed to sample the channels. After the 11.35 μs it asserts a signal telling State_B to move back to State_A.

3.2.2 DAC Timing

Part of the reasoning for having a sweep rate of 50 Hz was settling time requirements for the electronic components to settle out to an acceptable accuracy, about .01% accuracy. The electronics take approximately 20 μs to settle before sampling can begin. It also takes about 12 μs to complete the sampling process of the High and Low Gain channels. Originally it was desired to have 600 steps per sweep and take 33 μs per step, which is equivalent to a sweep rate of 50 Hz. However, the requirements were changed to better fit the PCM matrix developed by NSROC, and it was decided to keep the sweep rate at 50Hz and take 549 steps at about 32 μs per step.

The control of the DAC is accomplished with three signals. They are A0, A1, and WR. These signals control the writing of signals into two latches within the DAC. The method this design uses, to write a value into the DAC, is to make the latches transparent

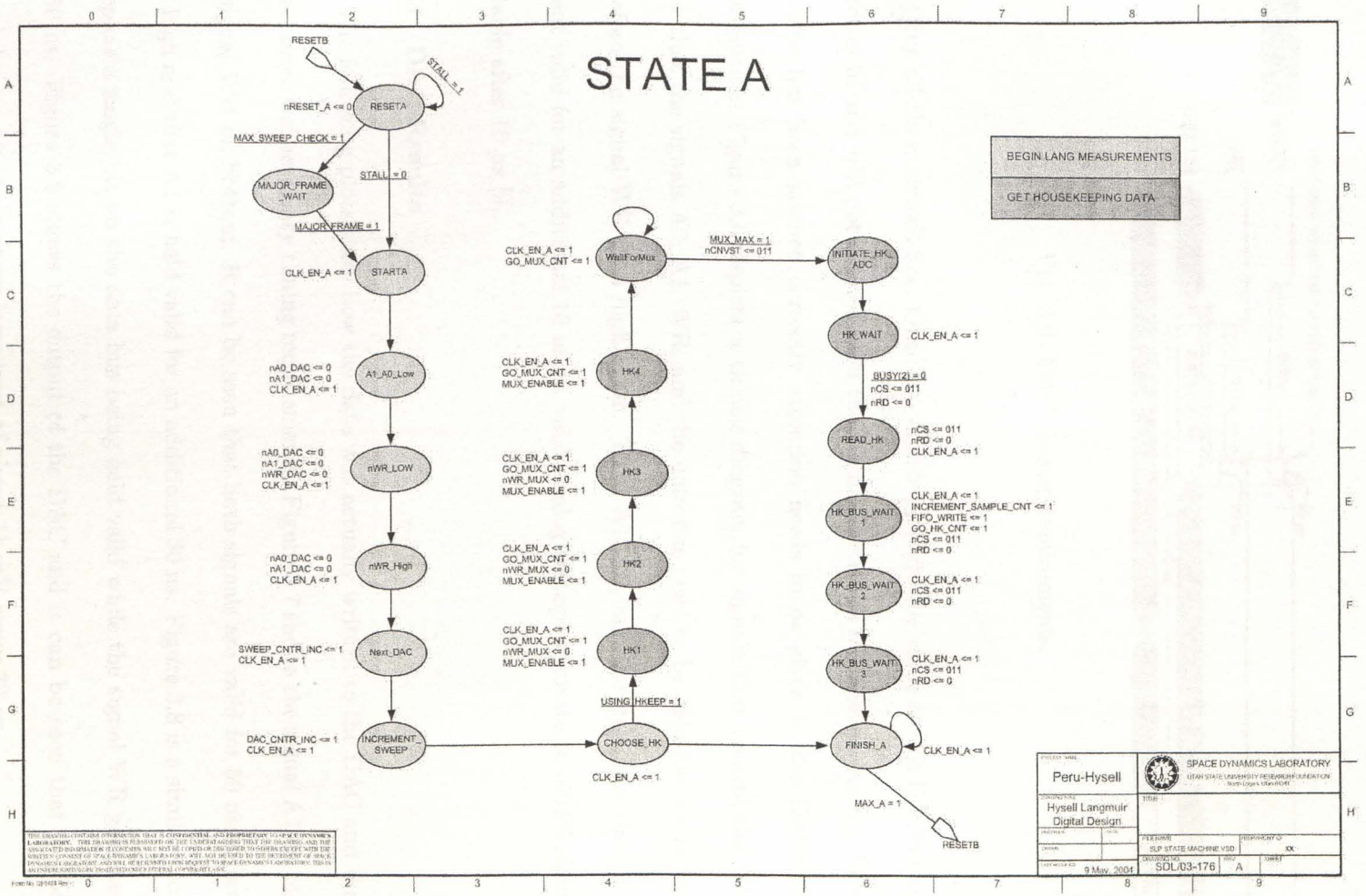

Fig. 3.5: STATE_A state diagram.

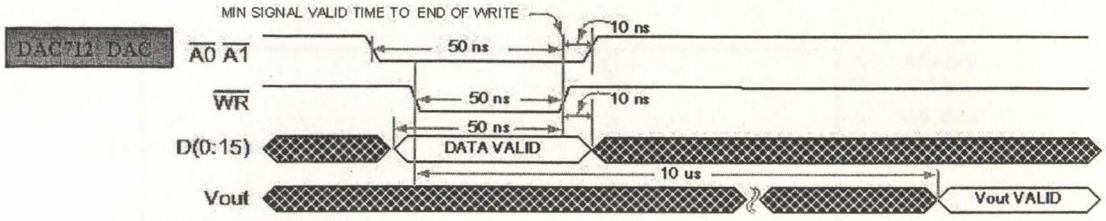

Fig. 3.6: DAC timing requirements.

by setting all three signals low. Once the signals go high, the value on the DAC's data bus is latched in and will not change until the latches are made transparent again. To ensure the value has been latched correctly attention needs to be given to the DAC's timing requirements. Figure 3.6 depicts a timing diagram for signals that control the DAC. It shows that the signals A0, A1, WR, and the data bus need to be held low for at least 50 ns before the signal WR goes high again. Then A1, A0, and the data bus then need to be kept valid for an additional 10 ns. A valid analog voltage accurate to .003% should be available after 10 μs [9].

3.2.3 DAC Results

The following plots show how the data was actually written to the DAC and that the signals meet the necessary timing requirements. Figure 3.7 shows the signal A1 on top and the signal WR on bottom. It can be seen that both signals are valid for 50 ns before WR goes high and that A1 is held valid for an additional 50 ns. Figure 3.8 is a similar plot, but it depicts a single bit on the data bus being held valid while the signal WR gets asserted for 50 ns. Figure 3.9 shows the output of the DAC and it can be seen that it works as desired, because it steps at an increment of about 11 mV every 32 μs.

3.3 ADC

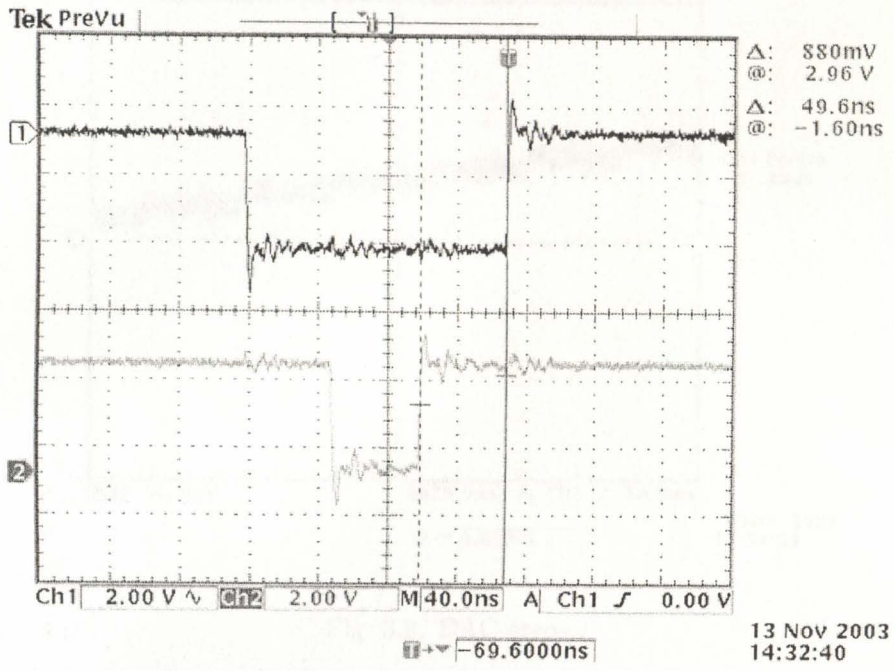

Fig. 3.7: A1 low and WR low widths.

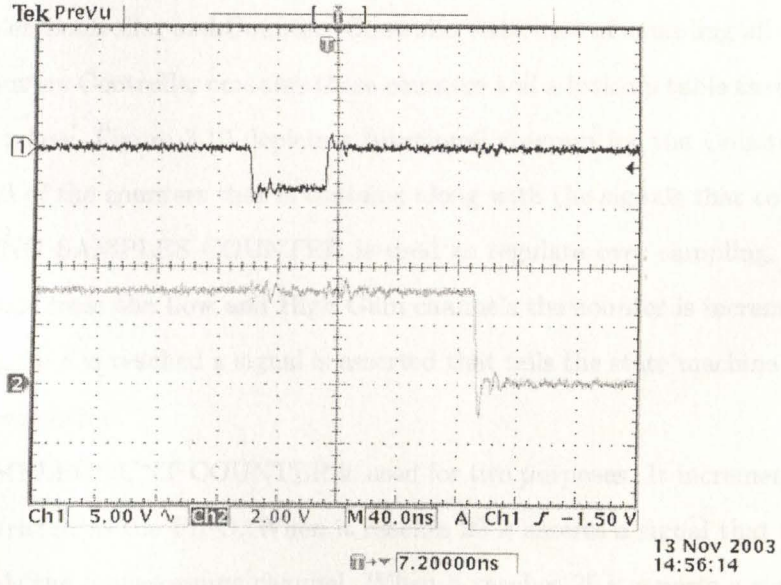

Fig. 3.8: A1 low and WR low widths.

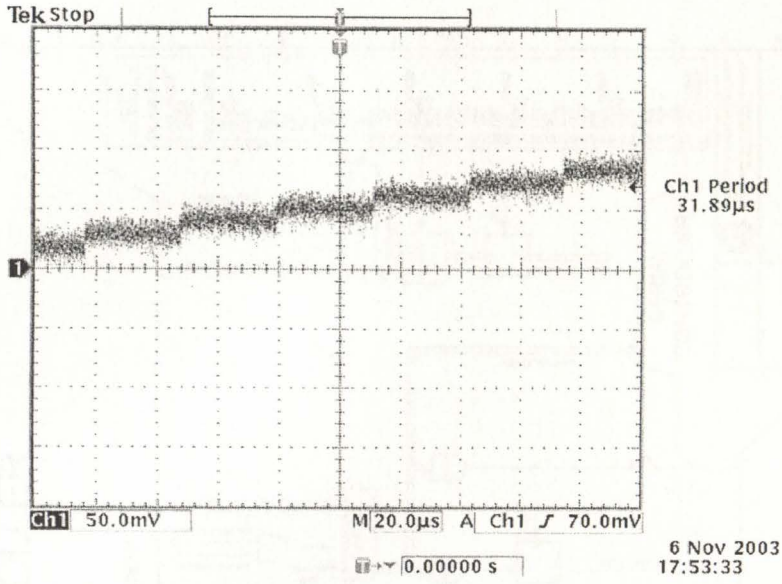

Fig. 3.9: DAC steps.

3.3.1 ADC Design

The next step in the control of the SLP is sampling the current from the probe, and sampling different housekeeping signals. State_B from the SLP State Machine, along with Sweep Sampler Controller and Counters Controller take care of sampling all the Channels.

The Counters Controller contains three counters and a look-up table that regulate how samples are taken. Figure 3.10 depicts a functional diagram for the Counters Controller block with all of the counters that it contains along with the signals that control them.

The LANG SAMPLES COUNTER is used to regulate over sampling. Each time a reading is made from the Low and High Gain channels the counter is incremented. When the max count of 8 is reached a signal is asserted that tells the state machine that the over sampling is completed.

The SAMPLECOUNT COUNTER is used for two purposes. It increments every time a sample is written to the FIFO. When it reaches 22 it asserts a signal that indicates it is time to sample the housekeeping channel. When it reaches 25 it asserts a stall signal that makes the state machine stop sampling. The stall signal stays asserted until it receives a signal from the telemetry encoder indicating that it is at a minor frame or at a major

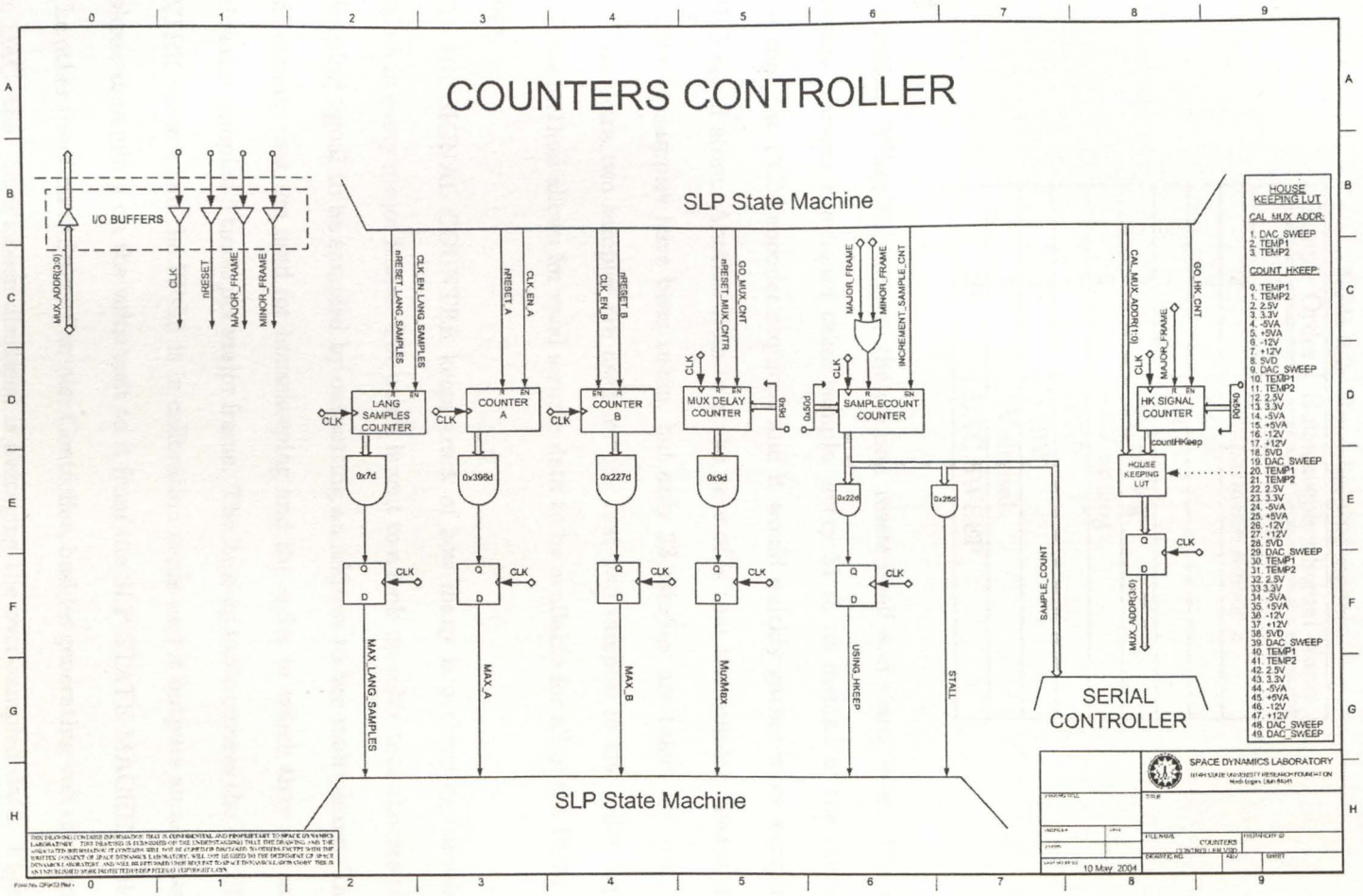

Fig. 3.10: Functional diagram for Counters Controller block.

Table 3.2: Housekeeping signals.

Sample Order	Housekeeping Signal Name
1	Temperature sensor 1
2	Temperature sensor 2
3	2.5 V
4	3.3 V
5	-5V analog
6	+5V analog
7	-12V
8	+12V
9	5V digital
10	DAC SWEEP

frame location. When this occurs the counter resets itself and starts over. The stalling is necessary because the board takes samples every 31.15 μs instead of the sample rate of 32 μs that the PCM encoder requires, and it would quickly gather more samples than the FIFO could store. Another item to make note of is that the counter asserts the stall signal after 25 samples have been taken, but only 23 samples are taken for every minor frame. The extra two samples are used for the first two samples in the following minor frame. This method allows for valid sample data to be available for all of the PCM matrix locations.

The HK SIGNAL COUNTER keeps track of how many housekeeping samples have been taken in every major frame. The count is sent to a look-up table that chooses the next housekeeping signal to be sampled by outputting an address to the multiplexer. Table 3.2 lists the signals that are used for housekeeping and the order in which they are sampled. Each signal is sampled 5 times per major frame. The look-up table ignores the HK SIGNAL COUNTER value when the FPGA is in calibration mode and it outputs an address to the multiplexer depending on the value sent to it from the SLP STATE MACHINE block.

The other function of Sweep Sampler Controller, besides generating voltage step values for the DAC, that hasn't been discussed is averaging the over-sampled data. Figure 3.11 shows how the samples are averaged. The High and Low Gain registers are reset to zero when sampling begins and then incoming data from the ADC_DATA bus are added

together. This is done eight times for each channel and then the last three bits of the resulting value is truncated. The truncation is equivalent to dividing by eight, giving the desired averaged result. An important fact to point out is that the averaged data sent to the Serial Controller isn't always valid. Only after being over-sampled eight times is it valid. The SLP State Machine keeps that in mind and only sends the signal to write the sample into the FIFO when the data is valid.

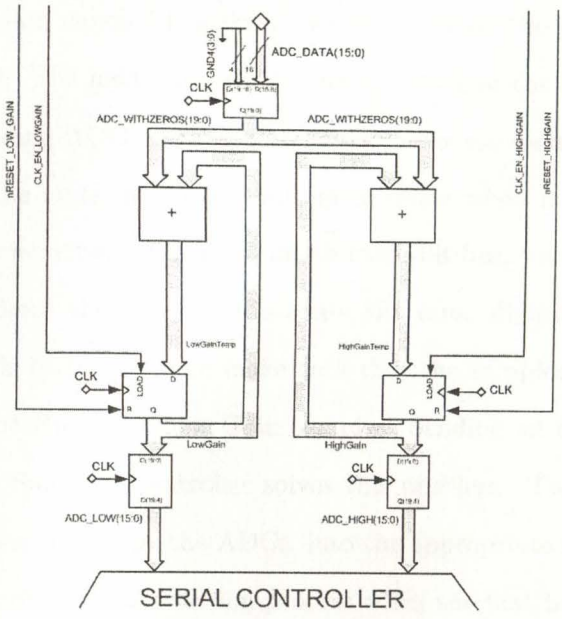

Fig. 3.11: Functional diagram for over sampling High and Low Gain channels.

Since most of the time in State_A is spent idly waiting, it makes sense to use some of that time to periodically acquire housekeeping signals fig. 3.5 shows the states to used to acquire data from the Housekeeping channel. The signal USING_HKEEP indicates that the next sample needed is a housekeeping signal. When USING_HKEEP is asserted the state machine deviates from its normal path and enables the multiplexer on the board to output the selected housekeeping channel and then enables the ADC to convert it to a digital signal. It then asserts signals to write the ADC output to the FIFO. Notice that no care is taken on deciding what channel to select on the multiplexer. This is because Counters Controller is already outputting the signals that choose the correct housekeeping

signal. The state machine just has to output the signals that enable the multiplexer and the ADC at the proper times.

State_B from the state machine represents the states that control the sampling of the High Gain and Low Gain channels of the probe. Figure 3.12 shows the state chart for STATE_B. It starts out resetting the registers that hold the accumulated values for the High and Low Gain channels. Both channels are sampled in parallel with each other. The sampling of the Low Gain channel is started first by enabling the ADC corresponding to the Low Gain channel. The next state starts the sampling of the High Gain channel by enabling its corresponding ADC. During a conversion process each ADC asserts a signal called BUSY, which the state machine monitors to know when it can start moving on. The three busy signals have been combined into a three-bit bus, were each bit corresponds to a different ADC. Since the three ADCs share the same digital output bus into the FPGA some care needs to be taken to make sure that the samples are read in correctly without intermixing the High and Low Gain samples. Sending an enable signal, for each channel, to the Sweep Sampler Controller solves this problem. This way it will register the signals, from the bus shared by the ADCs, into the appropriate register that has been enabled. For example, during a conversion process when the first bit in the bus "BUSY" goes low the state machine sends out the signal CLK_EN_HIGHGAIN to enable the High Gain register and to clock in the value on the ADC bus into that register. After both channels are registered the signal CLK_EN_LANG_SAMPLE is asserted to increment the mentioned LANG SAMPLES COUNTER to regulate the over sampling by eight. After sampling eight times the signal MAX_LANG_SAMPLES is asserted by LANG SAMPLE COUNTER and the state machine quits sampling and asserts signals that will write the averaged samples to the FIFO. It also asserts the signal INCREMENT_SAMPLE_CNT that SAMPLECOUNT COUNTER uses to keep track how many samples have been recorded. This helps in keeping track of when the next housekeeping sample needs to be taken.

3.3.2 ADC Timing

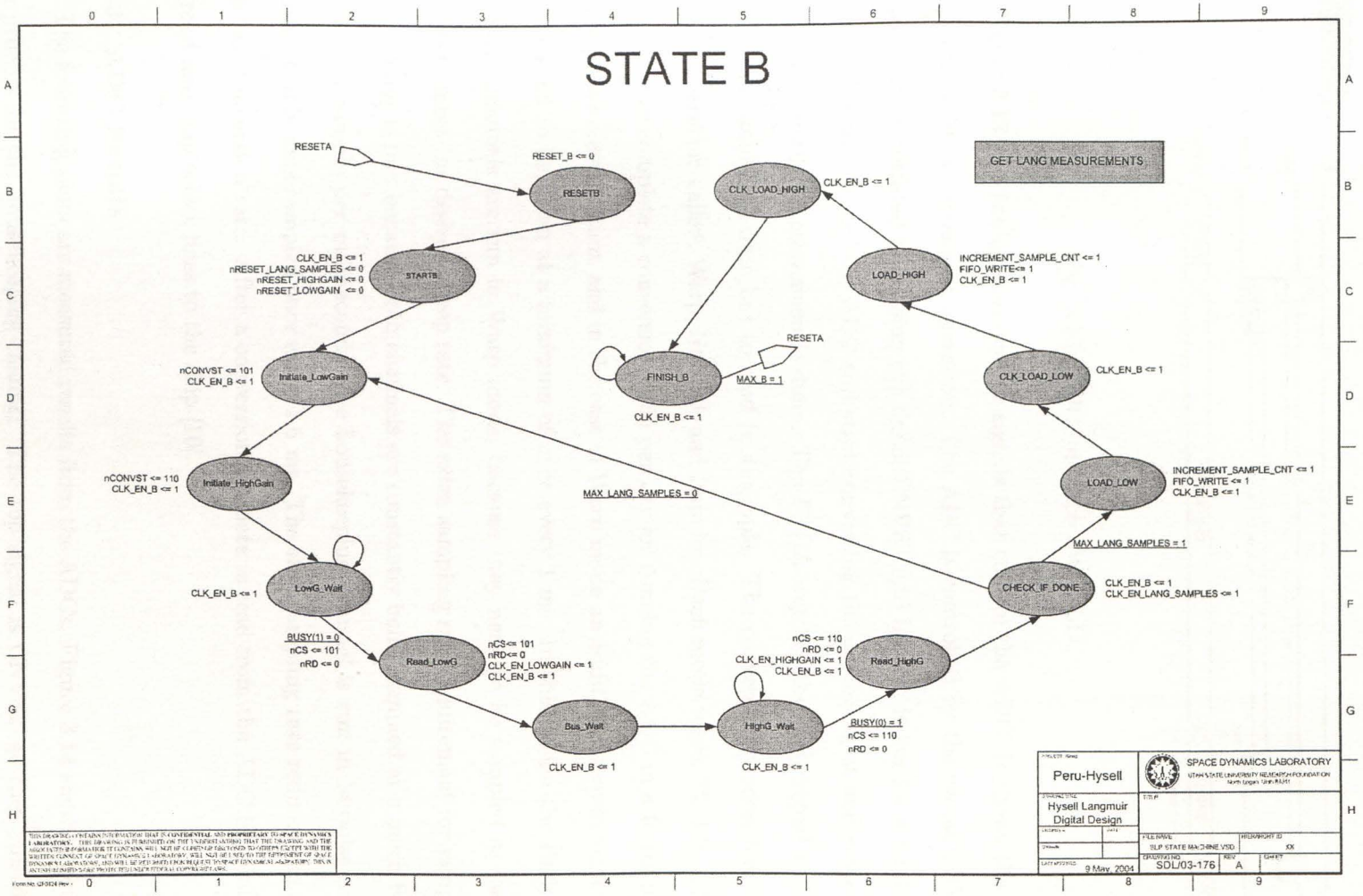

Fig. 3.12: STATE_B state diagram.

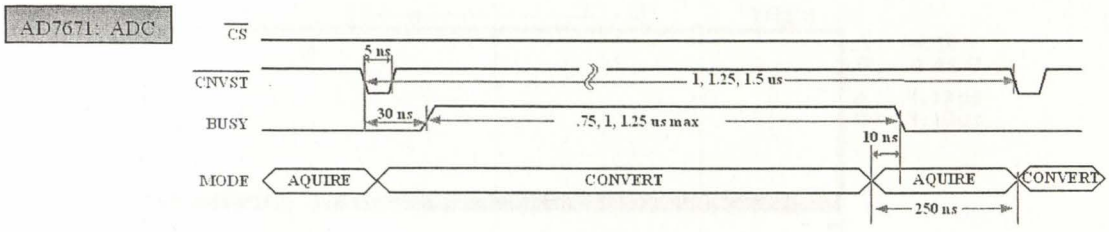

Fig. 3.13: ADC interface with PLD.

Figure 3.13 is a timing diagram for signals that control the ADC. It shows the necessary timing for a conversion operation. The ADC is controlled by the signal CNVST. A conversion is initiated by pulsing the signal CNVST and holding it low for a minimum of 5 ns. After this occurs the ADC will start converting the analog input and assert the signal BUSY until the conversion is done. The FPGA watches the busy signal to know when it can enable the data bus to read in a sample. The conversion process has three modes of operation called, Warp, Normal, and Impulse. Each mode takes .75, 1.0, 1.25 μs respectively to complete a conversion. The penalty for running the ADC in a faster mode is more power consumption, and in the case of Warp mode an additional requirement that samples need to be taken at a minimum of once every 1 ms. In this design the High and Low Gain channels are run in Warp mode, because they need to be sampled as fast as possible to meet the desired sweep rate. The extra sampling rate requirement for sampling in warp mode is met because both channels are constantly being sampled at a much faster rate than 1 sample per millisecond. The housekeeping channel is run in Impulse mode, because it only gets sampled once every 1.6 ms. The slow sampling rate reduces the need for a fast conversion rate. After a conversion the data is read from the ADC by enabling the read and chip select lines to the chip [10].

3.3.3 ADC Results

The following plots are measured results from the ADCs. Figure 3.14 shows a conversion process for the housekeeping channel. The top signal is CNVST getting pulsed low to start the conversion, and the bottom signal is the signal BUSY output from the ADC.

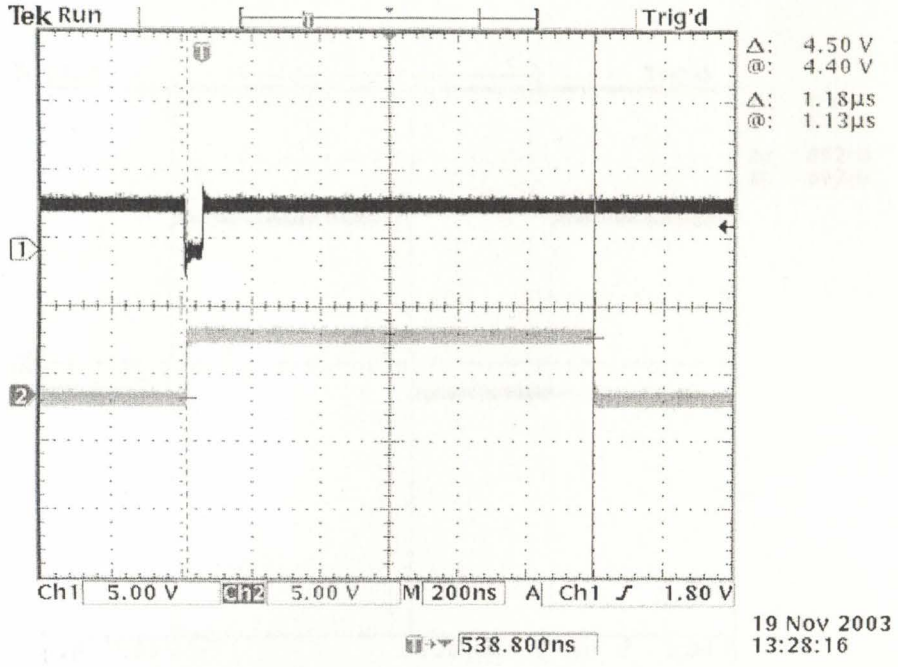

Fig. 3.14: CNVST (top) and BUSY (bottom) signals from ADC.

Notice that the length of the busy signal 1.18 μs indicates that the ADC is functioning in Impulse mode as expected. Figure 3.15 is a plot of the busy signal from the Low Gain channel. Also note that its length of 692 ns indicates that the ADC is functioning in Warp mode as expected. Figure 3.16 illustrates how the signals CS and RD work together. The top signal is the signal RD and the bottom one is CS for the High Gain channel. The FPGA was designed to read the Low Gain channel followed by the High Gain channel. By looking at the plot you can see that the signal RD, which is shared by all three ADCs, was asserted twice once to read the Low Gain channel then asserted again to read the High Gain channel. The signal CS, which corresponds to the High Gain channel, was only asserted when it was time to read the High Gain channel.

Figure 3.17 is what I consider the most interesting plot. It shows when sampling of the High Gain channel occurs. The signal that is stepping upwards is the High Gain channel and the signal that pulses is the BUSY signal from the ADC that corresponds the High Gain channel. It can be seen that sampling occurs 20 μs after a step occurs, which

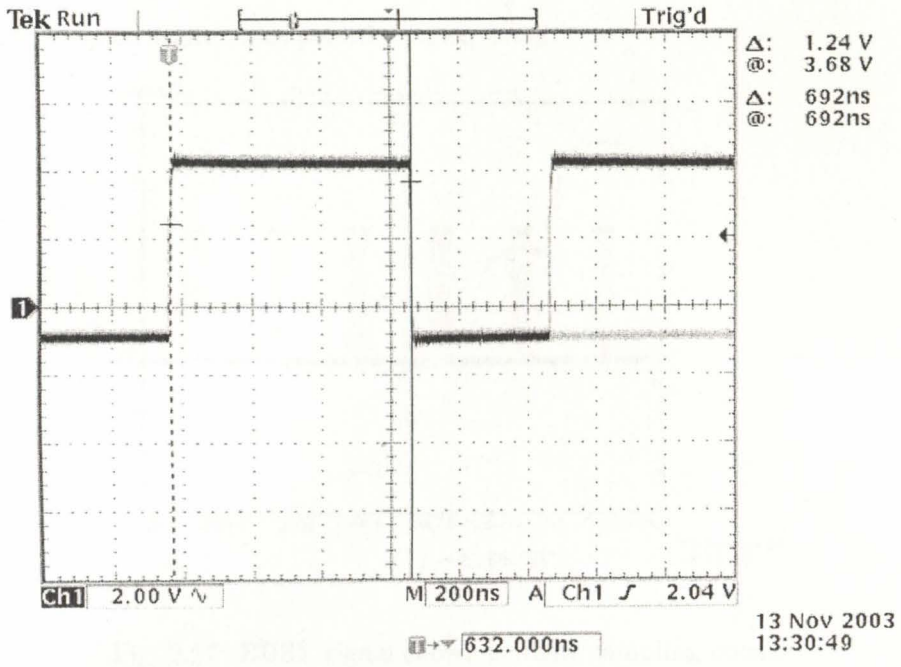

Fig. 3.15: Busy signal from Low Gain ADC.

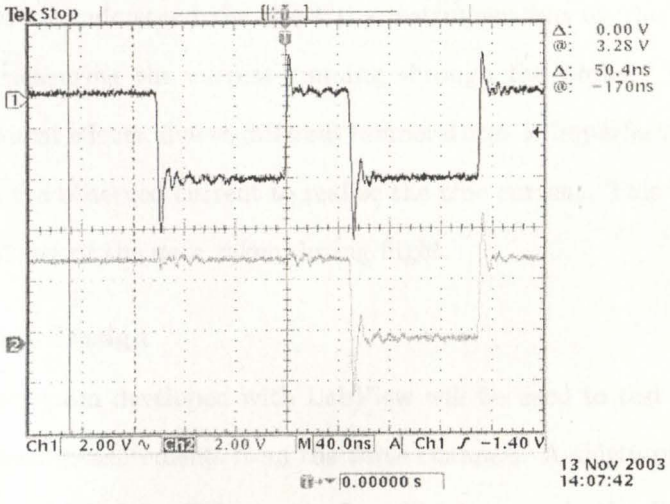

Fig. 3.16: RD (top) and CS (bottom) signals from ADC.

appears to be sufficient time for settling to occur. The plot also verifies that the channels are getting sampled eight times per step as expected.

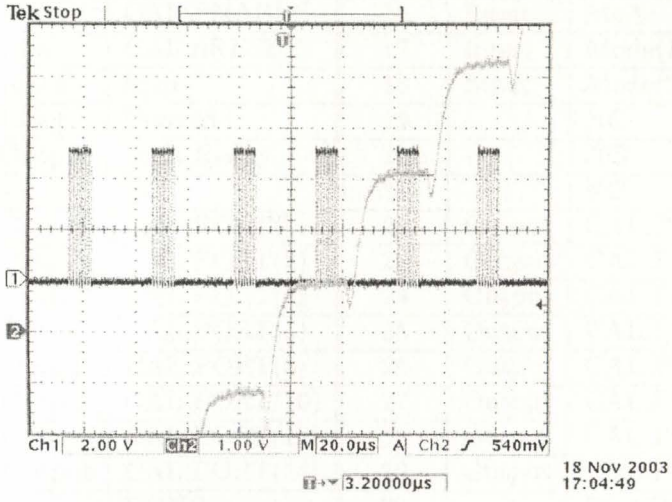

Fig. 3.17: BUSY signal showing where sampling occurs.

3.4 Calibration

The purpose of calibration is to measure how the instrument works under varying conditions in order to understand the effect the instrument has on the readings. In this program we are measuring the current running through the probe. We would like to remove any instrument effects, due to different temperatures or imperfections in the system components, from the observed current to realize the true current. This will enable better understanding and use of the data taken during flight.

3.4.1 Calibration Design

A computer program developed with LabView will be used to test different areas of the board and collect measurements from the three channels. A thirty-pin connector with a pin out shown in Table 3.3 will be used. In calibration mode the PC will be able to control voltage levels output by the DAC, and read samples from the High Gain channel, Low Gain channel, the DAC output voltage, and both temperature sensor. The three-bit

Table 3.3: Pin out for calibration port.

Calibration Pin Out					
Pin #	IO	Signal Name	Pin #	IO	Signal Name
1	Input	CAL_ENABLE	16	Input	Mode
2	Input	CAL_nRESET	17	Input	Mode(1)
3	Input	Step	18	Input	Mode(2)
4	Input	Convert	19	-	NC
5	Output	Data_Ready	20	-	NC
6	-	NC	21	-	NC
7	Output	CAL_PORT(0)	22	Output	CAL_PORT(1)
8	Output	CAL_PORT(2)	23	Output	CAL_PORT(3)
9	Output	CAL_PORT(4)	24	Output	CAL_PORT(5)
10	Output	CAL_PORT(6)	25	Output	CAL_PORT(7)
11	Output	CAL_PORT(8)	26	Output	CAL_PORT(9)
12	Output	CAL_PORT(10)	27	Output	CAL_PORT(11)
13	Output	CAL_PORT(12)	28	Output	CAL_PORT(13)
14	Output	CAL_PORT(14)	29	Output	CAL_PORT(15)
15	-	DGND	30	-	DGND (0)

Table 3.4: List of different functions chosen by bus MODE.

Mode		
Value	Mode Name	Function
000	DAC Sweep	Increment voltage on DAC by one step and read output voltage
001	ADC Low Gain	Read Low-Gain channel
010	ADC High Gain	Read High-Gain channel
011	ADC Temp1	Read Temperature sensor #1
100	ADC Temp2	Read Temperature sensor #2

bus named MODE selects the channel to be sampled. Table 3.4 shows the values of MODE that are necessary to select each channel.

Figure 3.18 shows a simplified version of the states that SLP State Machine follows when in calibration mode.

After calibration mode is selected the state machine waits in the "Go Convert" state until the signal CONVERT goes high. Then depending on the value of MODE a sample is taken from one of the channels. If DAC_SWEEP is selected the FPGA will increment the step value for the voltage, and then read back the corresponding voltage value from the housekeeping channel's ADC. If either High Gain or Low Gain are selected the value output to the PC is the sample value over-sampled eight times and averaged. This will

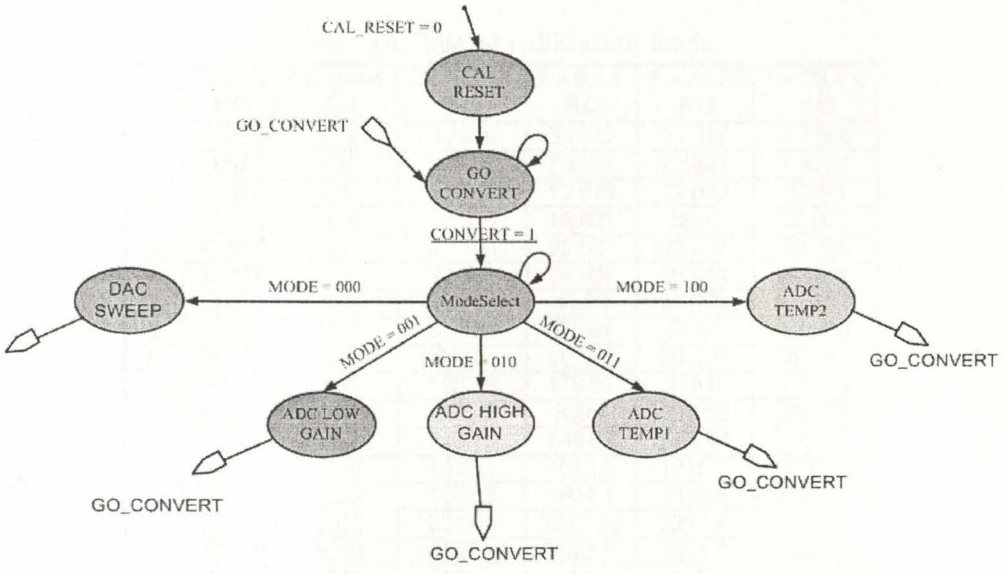

Fig. 3.18: Example of how a sample is chosen while in calibration.

allow for similar comparison with the measured values from the channels when the board is in RS422 mode, because those channels are also over-sampled eight times. When either of the temperature sensors is selected the output is a single sample of the temperature.

Twenty resistive loads where selected to help simulate the range of currents that will be seen in the ionosphere. A load will be attached to the board in place of the Langmuir boom, and a voltmeter will measure the voltage across the load at all voltage steps of the sweep. This will be repeated for the different loads. Table 3.5 shows the loads that were selected for calibration. The Low Gain channel was designed to measure currents from the range of 191.6 μA with 5.85 nA precision. Loads 1-10 are used to read Low Gain channel currents within the supported range without causing the op-amps to saturate. The High Gain channel was designed to measure currents from the range of 1.897 μA with 57.9 pA precision. The Loads 11-20 are used in a similar way to the Low Gain channel loads, except they were chosen to measure currents within the range that the High Gain channel could measure.

The voltmeter measurement is to aid in giving independent readings of the current through the load. By knowing the value of the resistor used and the voltage read by the

Table 3.5: List of calibration loads.

ID	Theory KΩ	Assigned KΩ	T = -0.2° C KΩ	T = 0.9 C KΩ	T = 23.2C KΩ	T = 39.4 C KΩ
1	3.7495	3.74	3.7476	3.7349	3.7348	3.7354
2	7.6572	7.5	7.4684	7.4571	7.4641	7.4707
3	11.91	12	12.12	12.018	12.013	12.03
4	16.759	16.9	16.925	16.801	16.81	16.821
5	22.597	22	22.166	22.051	22.057	22.065
6	30.096	30	30.101	29.997	30.011	30.029
7	40.578	41.2	41.408	41.29	41.279	41.287
8	57.103	57.6	57.738	57.654	57.66	57.663
9	88.814	90.9	91.54	90.589	90.597	90.614
10	181.38	180	180.39	179.52	179.6	179.75
11	378.83	390	390.3	389.7	389.83	389.89
12	773.66	750	746.15	745.62	745.49	745.4
13	1203.3	1020	1016.3	1015.6	1015.7	1015.7
14	1693.3	1800	1803.3	1804.4	1803.8	1803.7
15	2283.1	2200	2214.1	2213.35	2211.7	2211
16	3040.8	3010	30274	3026.7	3024.8	3023.6
17	4099.9	3900	4186.5	4185	4110	4071
18	5769.5	5600	6007	5997	5892	5834
19	8973.4	8200	8423.5	8408	8238	8144
20	18326	18000	18500	18500	18100	17900

voltmeter the current can be calculated. This current measurement will be compared to the current readings from the High and Low Gain channels. This procedure will also be done over different temperatures so the changes due to temperature can be parameterized. Another test will record 100 samples of the High and Low Gain channels per voltage step. This test will be helpful in showing how much noise is in the system, because it will show how much samples differ for a given voltage step.

Figure 3.19 illustrates how the system is to be set up for calibration [11]. A PC will be connected to a voltmeter and the calibration port of the probe. A GUI will then be developed that will control the sweeping of voltage on the probe and record measurements from the High and Low Gain channels, the voltage level on the DAC, the temperature sensors, and the voltage read by the voltmeter. The GUI will make these measurements every voltage step in one sweep of the probe. This will be repeated for each of the resistors. Figure 3.20 shows an image of the LabView GUI. The GUI allows the user to choose which load is being used, the temperature the test is being run at, and the serial number of the probe since two probes are being built and tested for flight. That information will allow

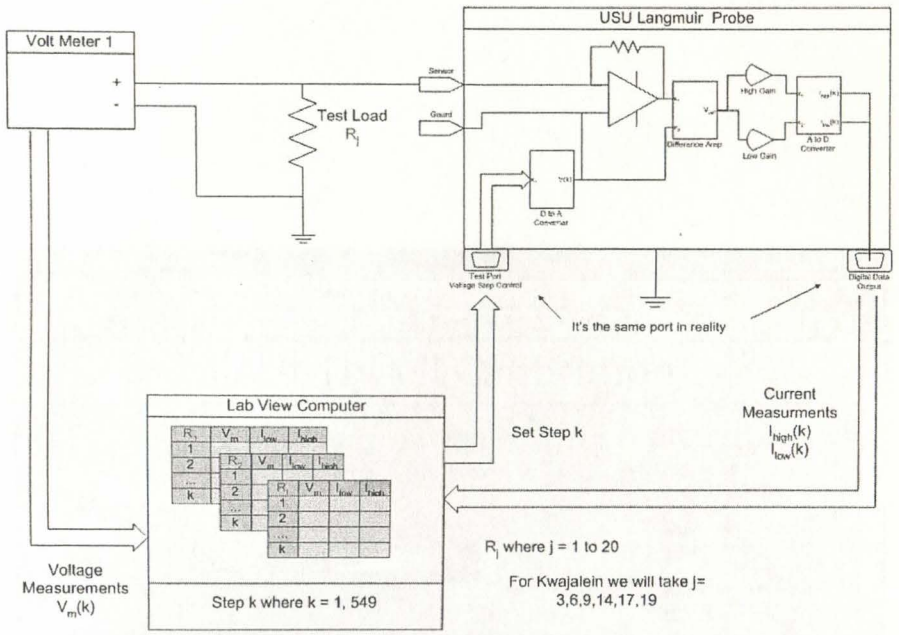

Fig. 3.19: Example of how a sample is chosen while in calibration.

for the saved files to be distinguished one from the other when the data is analyzed. The GUI will allow for an entire sweep to be run and recorded automatically, or the user can manually select a voltage in the sweep to run to and then look at the corresponding outputs on the screen.

3.4.2 Calibration Results

The following plots show the results of various tests run during calibration. Figure 3.21 shows the noise on the High Gain channel with load 17, which is a 3.8 MΩ resistor. The plot depicts that there are about 30 counts of noise in the High Gain channel. This is equal to about 1.7 nA of noise in the High Gain since each count represents 57.9 pA. The noise in the High Gain channel is an acceptable amount and it is close to the desired amount of 1 nA. Figure 3.22 is another plot of the High Gain channel with 100 samples per step. This plot looks very noisy because each individual sample can't be clearly distinguished with its corresponding step value, but the noise is with in its acceptable range. Each voltage step consists of 36 counts, and since there are about 30 counts of noise in the High Gain channel it can be hard to distinguish one step from the other.

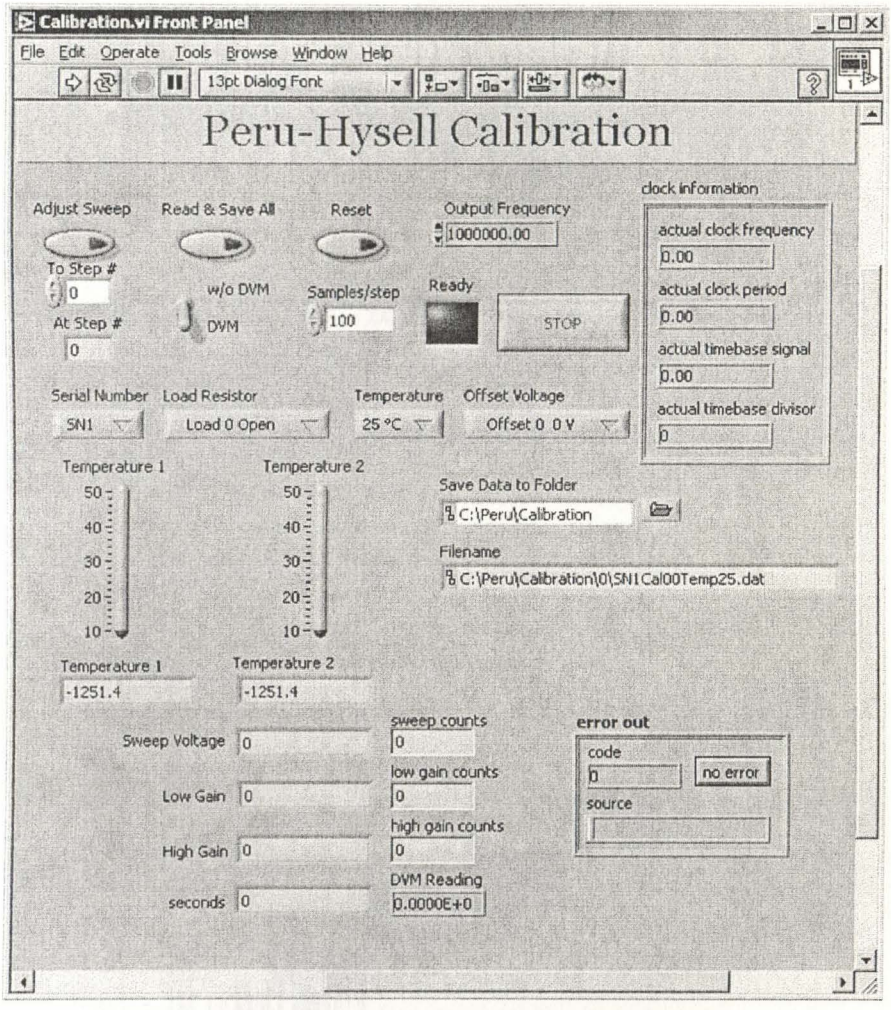

Fig. 3.20: Calibration GUI.

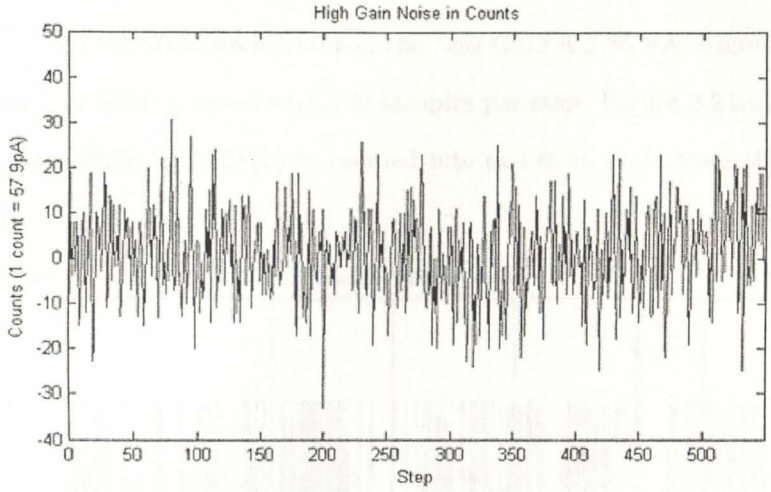

Fig. 3.21: High Gain noise in counts.

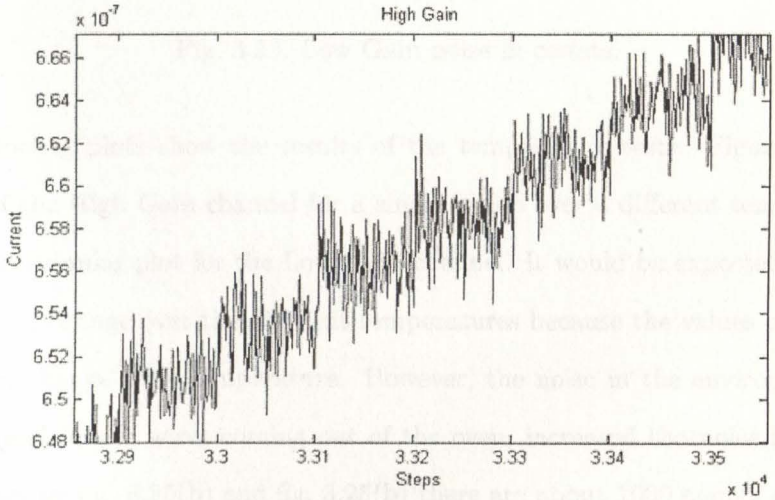

Fig. 3.22: High Gain channel with 100 samples per step.

The Low Gain channel is much cleaner but actually has more noise. Figure 3.23 shows that there are only two counts of noise in the Low Gain channel. The two counts of noise represent about 11.7 nA since each count in the Low Gain is 5.85 nA. Figure 3.24 contains two plots of the Low Gain channel with 100 samples per step. Figure 3.24(a) shows several steps of the sweep while fig. 3.24(b) is zoomed into one step, and shows that it varies by about 2 counts.

Fig. 3.23: Low Gain noise in counts.

The following plots show the results of the temperature tests. Figure 3.25(a) contains plots of the High Gain channel for a single sweep over 4 different temperatures and fig. 3.26(a) is a similar plot for the Low Gain channel. It would be expected that slope of the plots would change over the different temperatures because the values of the resistors in the system change with temperature. However, the noise in the environment, caused by the oven and longer wires coming out of the oven, increased the noise in the output. As can be seen by fig. 3.25(b) and fig. 3.25(b) there are about 1000 counts of noise in the High Gain channel and 20 counts of noise in the Low Gain channel. We can conclude that the temperature drift must be within the noise margin, which explains why we can't see any differences in the temperature tests.

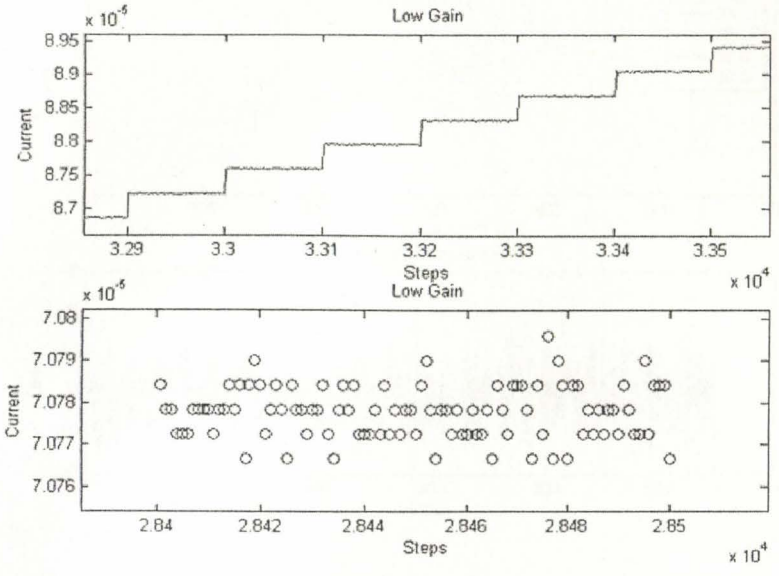

Fig. 3.24: (a) Low Gain with 100 samples per step. (b) Zoomed view of a single step.

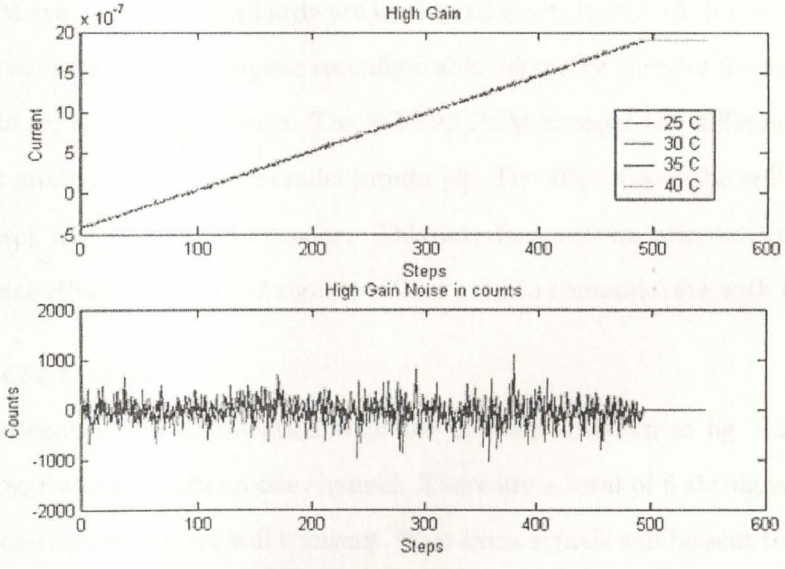

Fig. 3.25: (a) High Gain temperature sweep. (b) High Gain noise in counts.

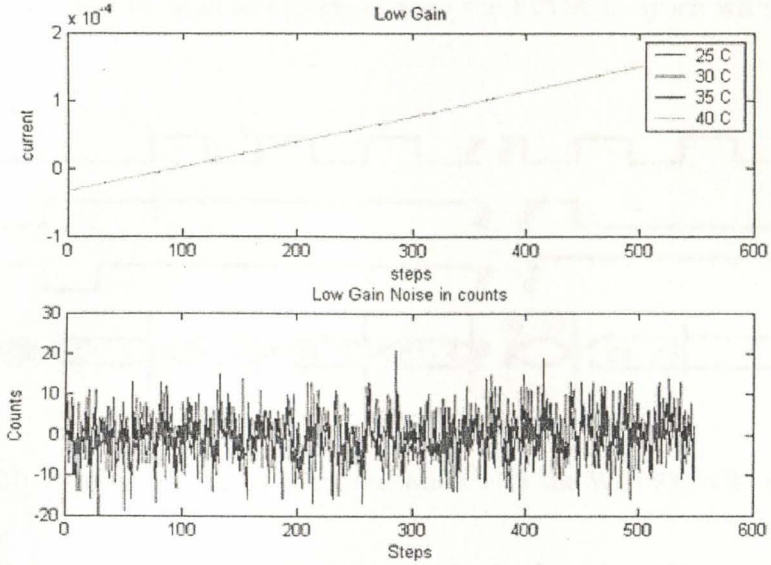

Fig. 3.26: (a) Low Gain temperature sweep. (b) Low Gain noise in counts.

3.5 NASA Serial

While in RS422 mode the FPGA will communicate with a WFF93 PCM system. The WFF93 PCM system is encoder hardware used to support the NASA Sounding Rocket Program. It serves as a general-purpose reconfigurable telemetry encoder for use when system requirements are subject to change. The WFF93 PCM encoder has different modules that can support analog, serial, and parallel inputs [8]. The High Gainrobe will interface with the serial deck of the WFF93 encoder. This interface will be referred to as the RS422 interface, since RS422 differential signals will be used to communicate with the serial deck.

3.5.1 RS422 Design

The protocol for communication with the encoder is shown in fig. 3.27. It requires four signals to transmit data on one channel. There are a total of 6 six signals that we will receive and one signal that we will transmit. Two extra signals will be sent to indicate when the encoder is at a major frame or minor frame location. The signal MAJOR_FRAME gets asserted high when the PCM matrix is in the last cell of the matrix. The signal MINOR_FRAME gets asserted high when PCM matrix is on the last cell of each row of

the matrix. They will be used as signals to keep the FPGA in synch with the telemetry encoder.

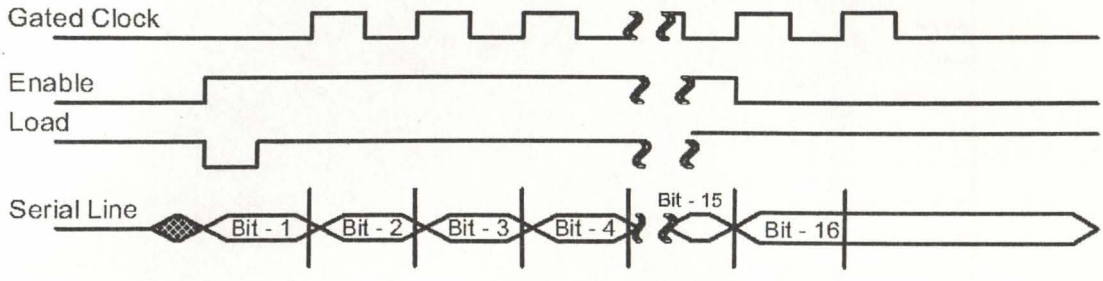

Fig. 3.27: Protocol for serial communication with the WFF93 PCM encoder.

The word that is to be transmitted needs to be loaded when either nLOAD goes low or when ENABLE goes high. Each bit should then be shifted on the rising edge of each gated clock. The last bit however should not be shifted after the final gated clock rising edge, rather it should be held until the next word gets loaded. There is a different protocol when two contiguous words are transmitted, but since the PCM matrix does not contain to contiguous words we will not need to worry about that protocol.

The PCM matrix in fig. 2.3 is 50X50 words long with each word containing 16 bits. The matrix runs at a rate of 50 major frames per second, and each bit will be transferred into the matrix at a rate of 2 Mbits/sec. Notice that the voltage sweep and the major frame rates are the same at 50 HZ. This will allow for all the data for an entire sweep to fit within one major frame. The matrix allots the Langmuir Probe 23 columns. One column will be used for housekeeping data, and the High and Low Gain channels will split the remaining 22 columns giving them 11 each. This setup will allow for 550 samples per major frame. However, after much consideration it was decided that we could use only 549 samples per major frame for each channel and replace the first samples of the High and Low Gains with a Major Frame ID. Which is also the reason a sweep contains 549 steps. The change was made due to the reason that the sweeping of the probe was designed to synchronize with the telemetry encoder by resetting itself at the end of a major frame.

Table 3.6: Data rate requirements.

Signal Name	Bits	Min Rate Hz	Max Rate Hz	Actual Rate Hz
LANG_LOW	16	27,450	27,450	27,450
LANG_HIGH	16	27,450	27,450	27,450
TEMP1	16	.1	10,000	250
TEMP2	16	.1	10,000	250
2.5V	16	.1	10,000	250
3.3V	16	.1	10,000	250
-5VA	16	.1	10,000	250
+5VA	16	.1	10,000	250
-12V	16	.1	10,000	250
+12V	16	.1	10,000	250
5VD	16	.1	10,000	250
DAC_SWEEP	16	300	10,000	300
MAJOR Frame ID	16	50	10,000	100

After performing the reset it would be difficult to have valid data ready for the first two locations in the PCM matrix. Placing a count of the number of sweeps ran in the first two locations eliminated the complexity of having data ready to output shortly after a reset.

The data rates to meet the science requirements are given in Table 3.6. As can be seen, by the table, the PCM matrix easily supports recording data at sufficient rates.

The Serial Controller block in the digital design is used for communication with the NASA's telemetry encoder. It mainly consists of a FIFO, a multiplexer, a look-up table, and a shift register. Figure 3.28 help gives a visual aid of how the look-up table and the multiplexer work together. The look-up table provides an address to the multiplexer and selects a channel to output into a FIFO. Two signals and a count control the look-up table. The signal vector SAMPLE_COUNT indicates the number samples taken in a minor frame. When the count is even the look-up table selects the High Gain channel from the multiplexer, and when the count is odd it selects the Low gain channel, except on the 23rd count then it selects the housekeeping channel. There are two signals that cause the look-up table to select different channels than the default ones selected by SAMPLE_COUNT. When the MAJOR_FRAME signal from the telemetry encoder is asserted high then the value from a 16-bit counter is selected from the mux. This allows us to output, at the

beginning of each PCM matrix major frame, the number of major frames or sweeps that have been taken. The storing of the major frame count number at the beginning of each matrix also serves the purpose of allowing time to begin and finish making a new sample of the High and Low Gain channels before they need to be output. The other signal that can make exceptions to the output of the look-up table is TEST_PATTERN. When this signal is asserted the constant hex word DEAF is output in place of the Low Gain channel and the constant hex word BABE is output in place of the High Gain channel. The housekeeping channel still gets output as normal. The main purpose for outputting predefined values is for debugging the serial communication link. It is a quick and easy way to see if received values in the telemetry encoder match what was output from the SLP.

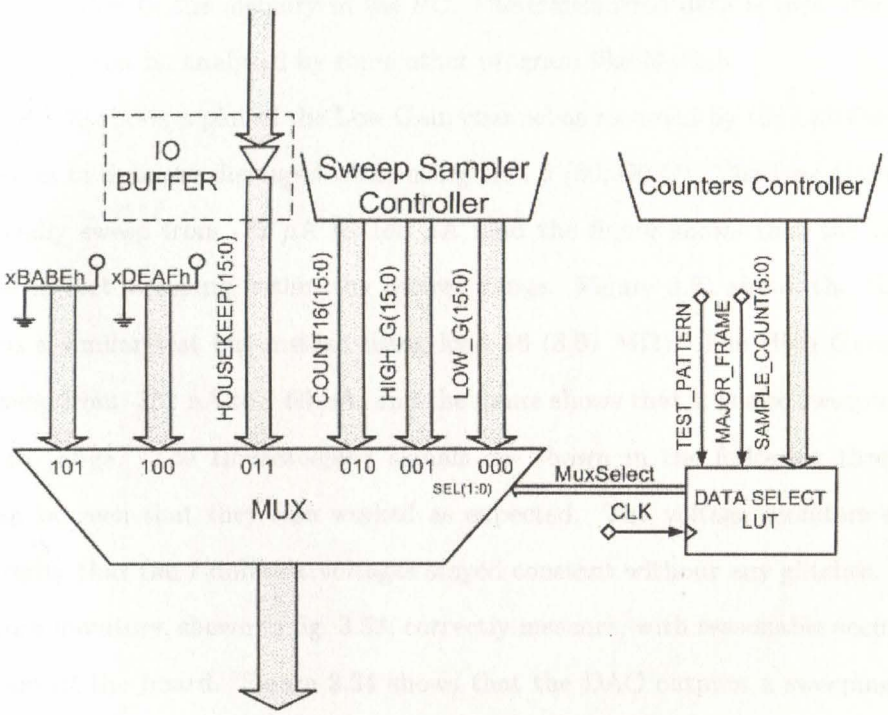

Fig. 3.28: Look-up table and multiplexer in the Serial Controller block.

After a signal is selected from the multiplexer it goes to the FIFO and waits for a write signal from the SLP State Machine. In the case of the 16-bit counter two write signals are generated within the Serial Controller and the value is written into the FIFO twice, so it can take the place of one Low Gain sample and one High Gain sample. Samples

are read from the FIFO with signals coming from the telemetry encoder. A pulsed signal nLOAD reads a sample from the FIFO and sends it into a shift register. As long as the signal ENABLE is held high the shift register outputs the sample bit-by-bit. The shift register, which is driven by a gated clock sent by the telemetry encoder, outputs the most significant bit first.

3.5.2 RS422 Interface Results

In order to test the functionality of the RS422 interface a GUI programmed with LabView was made to simulate the WFF93 PCM encoder. The GUI allows the user to choose how many major frames to record. The number of major frames that can be recorded is limited to the memory in the PC. The transmitted data is then stored into a binary file that can be analyzed by some other program like Matlab.

Figure 3.30 shows a plot of the Low Gain channel as recorded by the LabView GUI. 9 major frames of data are displayed while using load 6 (30,000 Ω). The Low Gain channel should ideally sweep from -33 μA to 166 μA, and the figure shows that the Low Gain channel is in fact sweeping within the desired range. Figure 3.31 shows the High Gain channel in a similar test but instead using load 16 (3.01 MΩ). The High Gain channel should sweep from -332 nA to 1.66 μA, and the figure shows that it is also sweeping within the desired range. The Housekeeping signals are shown in the following three figures and it can be seen that they also worked as expected. The voltage monitors shown in fig. 3.32 verify that the 7 different voltages stayed constant without any glitches. The two temperature monitors, shown in fig. 3.33, correctly measure, with reasonable accuracy, the temperature of the board. Figure 3.34 shows that the DAC outputs a sweeping voltage ranging from -1V to 5V as required.

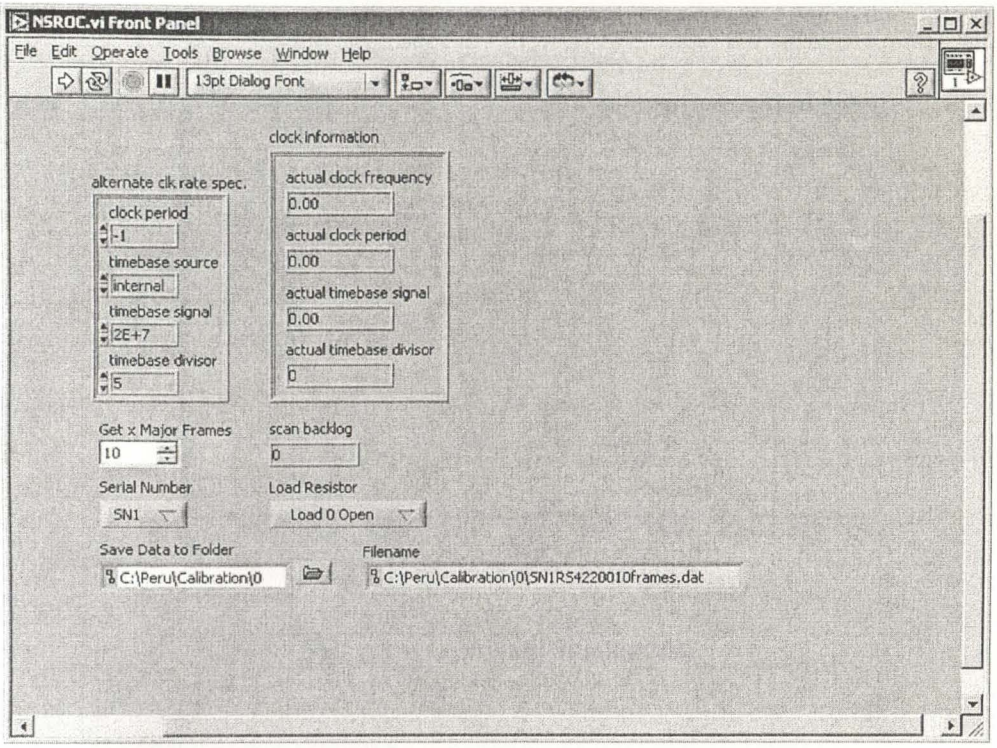

Fig. 3.29: LabView GUI for RS422 interface.

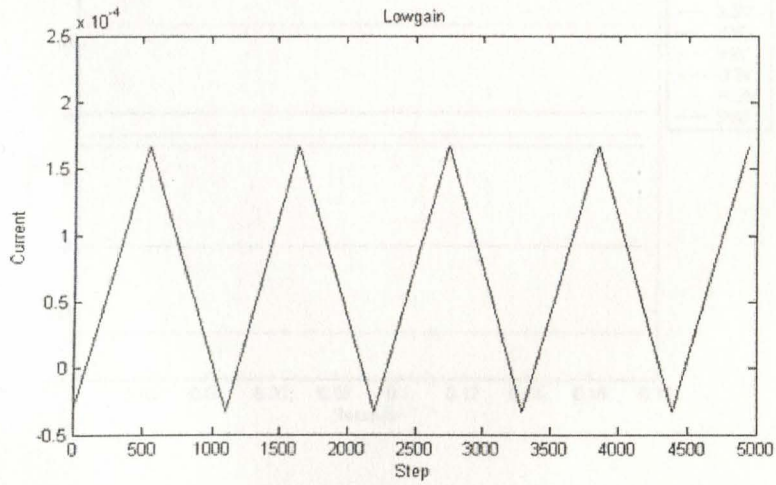

Fig. 3.30: Low Gain RS422 data.

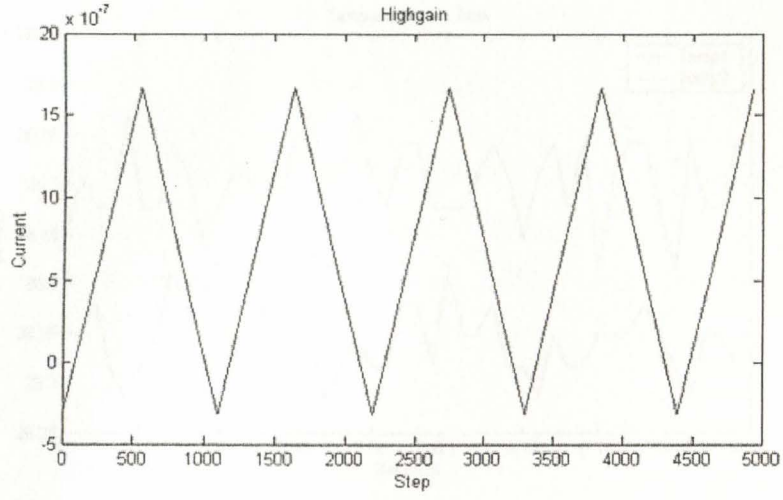

Fig. 3.31: Low Gain RS422 data.

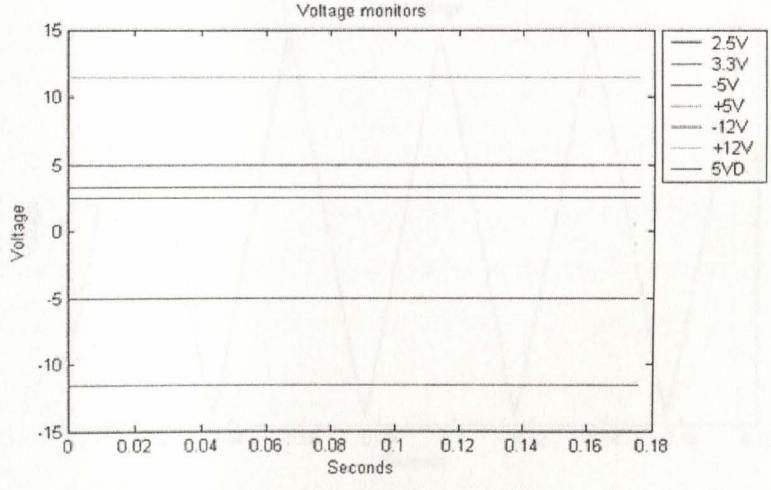

Fig. 3.32: Voltage monitors.

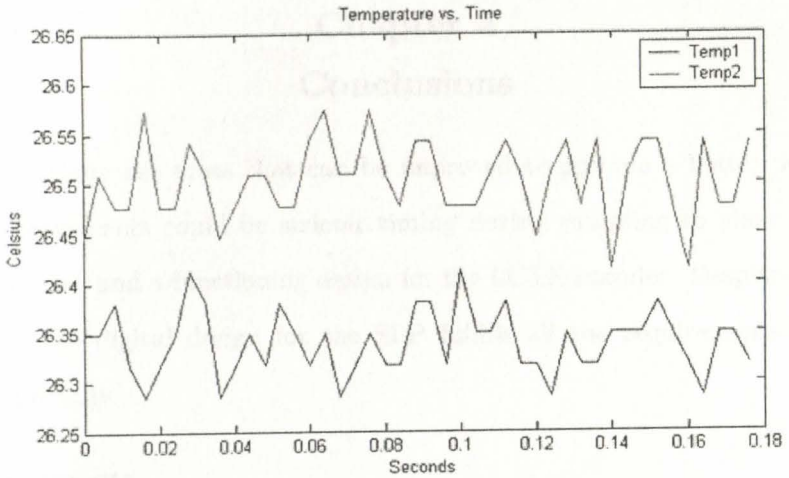

Fig. 3.33: Temperature monitors.

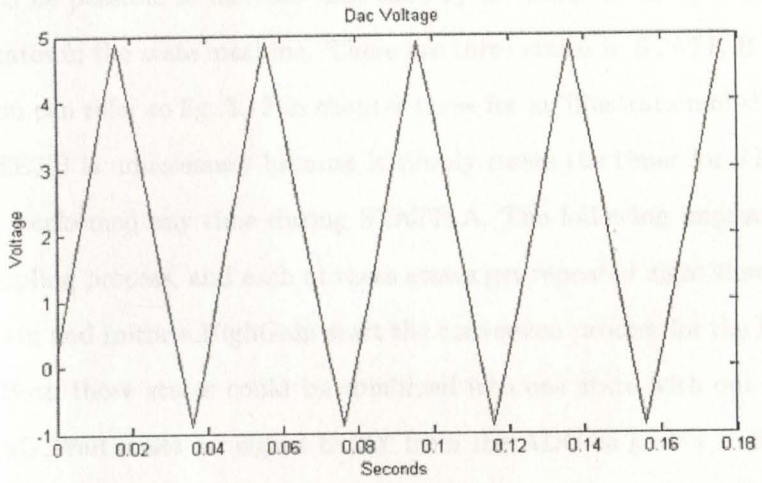

Fig. 3.34: Sweeping voltage output.

Chapter 4
Conclusions

As always there are areas that can be improved to provide a better overall system. A couple improvements could be stricter timing during sampling to allow more settling time for the probe, and a functioning design for the LCTE encoder. Despite these possible improvements the digital design for the SLP fulfills all the requirements specified and functions beautifully.

4.1 Improvements

4.1.1 Timing

One improvement could be added by giving the probe more time to settle out after each voltage step is taken. This may not reduce the noise that much, but I would like to see the results of this change. Currently the probe is given 19.8 μs to settle before samples are taken. It would be possible to increase that time by an extra .85 μs by eliminating some unnecessary states in the state machine. There are three states in STATE_B that could be eliminated. You can refer to fig. 3.12 in chapter three for an illustration of the state chart. The state RESETB is unnecessary because it simply resets the timer for STATE_B. This reset could be performed any time during STATE_A. The following improvements occur during the sampling process, and each of these states get repeated eight times. The states Initiate_LowGain and Initiate_HighGain start the conversion process for the High and Low Gain ADCs. Both those states could be combined into one state with out any conflicts. The state HighG_Wait waits for signal BUSY from the ADC to go low indicating that a conversion is finished. This state is unnecessary, because at that point enough time has passed that the conversion process, for the High Gain channel, should always be finished. The elimination of these states would save 17 clock cycles giving an extra .85 μs that could be added on to the settling time, giving a total settling time of 20.65 μs.

Table 4.1: Proposed backplane signals.

Description	# of lines
Parallel address and data bus	**20**
Address/data (AD) lines	16
Output enable (OE) line	1
Address latch enable (ALE) line	1
Read/Write line	1
Wait line	1
Interrupt lines	**4-8**
PCM matrix signals	**5**
Major frame strobe	1
Minor frame strobe	1
Word strobe	1
Bit clock	1
2x bit clock	1
Addressable synchronous serial input	**6-8**
Clock	1
Enable	1
Load	1
Address lines	3-5
Total lines	**35-41**

4.1.2 LCTE Interface

Another area that stands in need of improvement is the interface with the LCTE. This interface was desired for future missions with the SLP, so not much attention was paid to it since it is not critical to this mission. Currently the SLP can physically interface with LCTE, via 42 lines in the backplane connector, but no firmware has been developed to allow both boards to communicate. Table 4.1 shows the signals that have been proposed to be used on the back plane [6]. If those signals were used the SLP could be adjusted to communicate with the LCTE by replacing the Serial Controller block with code that performs the functions shown in fig. 4.1. The only signals that would need to be used for communication from the backplane would be the Address/Data bus, the Address latch enable line, and the Read/Write line. With this method the Read/Write signal will tri-state the output from the multiplexer while an address is on the data bus. After the address is latched and the Read signal is asserted the output from the multiplexer can be output

onto the bus to be read by the LCTE. The SLP will also need the signals Major frame strobe, and Minor frame strobe to help keep the sweep in synch with the PCM matrix. Currently the WFF93 encoder is providing these signals, in order to support the change to the backplane connector the pin assignments for those signals will need to be changed.

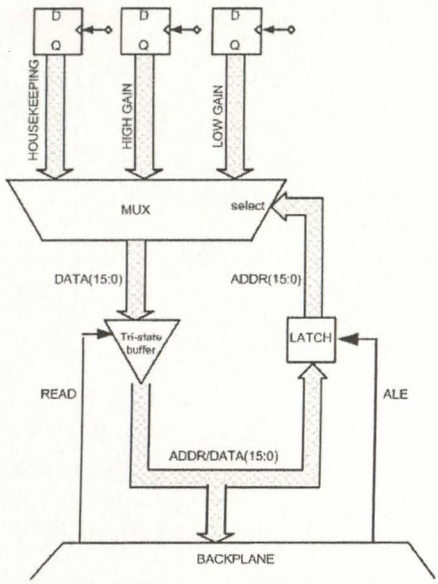

Fig. 4.1: Functional diagram for a backplane interface.

4.2 Instrument Functions

Testing has shown that the SLP can generate a sweeping voltage at the desired rate of 50 Hz, and that the SLP can sample the High and Low Gain channels during every step of the sweep, while giving adequate settling time for each channel. The tests have also shown that the over sampling of the High and Low Gain channels works properly. A calibration mode was also successfully implemented that helped us find ways to reduce noise in the system and also help us understand the effect the instrument has on the true readings. This will help in better analyzing ionospheric data gathered from flight. The RS422 serial link to a telemetry encoder was also successfully implemented and tested during integration at NSROC's facility in Wallops Island, VA. The testing verified that all sampled channels, including the housekeeping channel, were correctly transmitted to the PCM encoder. The

testing done during integration was done with actual hardware that will be used during flight. The integration showed a very positive achievement that indicates the SLP will be able to operate correctly during flight.

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Appendices

Appendix A
Kwajalein Langmuir Probe Calibration Test Plan

This document contains the calibration procedure for the Kwajalein Langmuir Probe as was outlined in a meeting held on 10 March 2004 at SDL. The attendees were: Charles M Swenson, Albert Hummel, Omar Morales, Tim Campbell and Aroh Barjatya.

1. Calibration of the loads. The loads that are being used for the instrument calibration need to be tested/calibrated under three different temperatures. In our case, we need them to be around 0 C, 20 C, and 40 C. This thermal testing will be done for each of the 20 loads, under the above temperature ranges and listed separately in a spreadsheet. The eventual task is to take this list and plot the different values of resistors against temperature to find their temperature coefficients.

2. Calibration of the instrument

   (a) Determine the resistance (Rmeter) of the Digital Volt Meter (DVM) being used. This has to be done via a simple experiment whereby we hook up a capacitor across the DVM leads, charge it to 5 Volts and then observe the decay time of the charge. This has to be done with 3-4 different capacitor values.

   (b) Calibration data: The test setup is illustrated in a drawing at the end of the document. The procedure will be in 9 parts as follows

      i. No Load (open) DVM Only

      ii. Load 3 + DVM

      iii. Load 6 + DVM

      iv. Load 9 + DVM

      v. Load 14 + DVM

      vi. Load 17 + DVM

vii. Load 19 + DVM

viii. Load 6; No DVM

ix. Load 17; No DVM

Procedure number i through vii will have one point for every voltage step in the calibration file. The file name will have the syntax snXcalYYtempZZ.dat, where X stands for the serial number of the board being calibrated, YY stands for the load being used, and ZZ stands for the ambient room temperature during testing. In the 'No Load' test procedure we put down the load number as 00. Procedure number viii and ix will not have the DVM hooked up and have 100 points for each voltage step. Having no DVM should minimize the noise and having 100 points per step will help us characterize the DAC performance and instrument noise. The file name should be snXcalYYtempZZ_noDVM.dat.

(c) Temperature Calibration of the instrument: During this test we will repeat procedure numbers iii through vi for an ambient temperature of 28 C, 30 C, 35 C and 40 C. The file name syntax will be just as before, and the temperature will be noted in the name of the data file created.

(d) RS422 Calibration: the purpose of this test is to show that no data is dropped and the system is able to perform well for a stretch of time. The data file should contain only binary format for 16 bit words. i. Long term test of about 30 to 60 seconds for Load 6 ii. Load test; 10 frames of data for each load (Load 0 to Load 20).

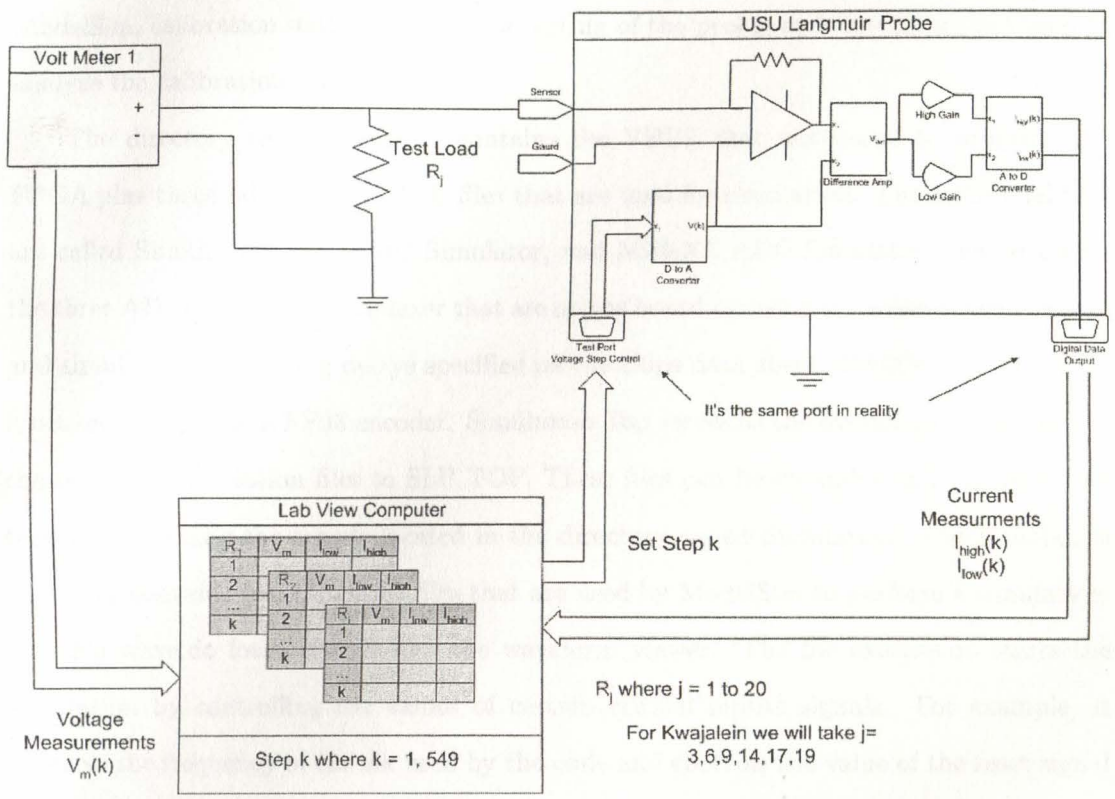

Fig. A.1: The calibration setup.

Appendix B
Source, Simulation, and Data CD

The included cd contains four directories that include the VHDL source code used to run the FPGA, command scripts used to simulate the functionality of the probe in ModelSim, calibration data taken during testing of the probe, and Matlab scripts used to analyze the calibration data.

The directory called "source" contains the VHDL that were used to program the FPGA plus three additional VHDL files that are used for simulation. The additional files are called Simulation Top, ADC Simulator, and NSROC. ADC Simulator tries to mimic the three ADCs and the multiplexer that are on the board by using the same control signals and simulating the timing delays specified on the chips data sheet. NSROC simulates the functionality of the WFF93 encoder. Simulation Top serves as the overall top level file and connects the simulation files to SLP TOP. These files can be compiled in ModelSim and tested by running the scripts located in the directory called Simulation. The Simulation directory contains two command files that are used by ModelSim to perform a simulation. The file wave.do loads signals into the waveform viewer. The file execute.do starts the simulation by controlling the values of certain critical inputs signals. For example, it controls the frequency of the clk used by the code and controls the value of the reset signal to the. Execute.do is the only file that needs to be loaded because it calls wave.do to set-up the waveform viewer.

The directory "data" contains all of the recorded calibration data. The naming convention for the calibration data file names are as outlined by the Calibration Test Plan in Appendix A. The Matlab scripts used to evaluate the data are located in the directory "Matlab".